Health Care Analysis

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Chandigarh University

Department of Computer Science and Engineering

April 2026

BONAFIDE CERTIFICATE

Certified that this project report "Business Intelligence Dashboard for HEALTH CARE Analytics using Microsoft Power BI" is the bonafide work of "Sachin Yadav(24BCS12785)" who carried out the project work under my/our supervision.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude and appreciation to all those who have contributed to the successful completion of this project. This project report on "Business Intelligence Dashboard for HEALTH CARE Analytics using Microsoft Power BI" has been a rewarding journey, and I am deeply indebted to several individuals for their guidance, support, and encouragement throughout this endeavor. First and foremost, I am extremely grateful to my project supervisor at the Department of Computer Science and Engineering, Chandigarh University, for their invaluable guidance, technical insights, and continuous motivation during the entire course of this project. Their expertise in data analytics and business intelligence proved instrumental in shaping the direction and quality of this work.

I extend my heartfelt thanks to the Head of the Department of Computer Science and Engineering for providing all necessary facilities and resources required to complete this project successfully. The department's commitment to fostering research and practical learning has been a constant source of inspiration.I am also grateful to all the faculty members of the Department of Computer Science and Engineering, Chandigarh University, whose lectures, workshops, and academic input have equipped me with the knowledge and skills required to undertake this project. I would like to acknowledge the contributions of Microsoft Corporation for providing comprehensive documentation and learning resources for Power BI, which proved indispensable throughout the development of this project.My sincere appreciation goes to my fellow students and peers who participated in feedback sessions and provided valuable suggestions for improving the dashboard design and functionality.I am deeply grateful to my family for their unwavering support, patience, and encouragement throughout my academic journey. Their belief in my abilities has been a constant source of strength and motivation.

Finally, I thank the Almighty for bestowing upon me the knowledge, wisdom, and perseverance to complete this project successfully.

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ABSTRACT

In the contemporary era of data-driven decision making, organizations across various industries are increasingly recognizing the critical importance of Business Intelligence (BI) tools for transforming raw data into actionable insights. This project presents the design, development, and implementation of a comprehensive Business Intelligence Dashboard for Sales and Operations Analytics using Microsoft Power BI — one of the most widely adopted BI platforms in the industry today.The primary objective of this project is to develop an interactive and visually compelling Power BI dashboard that enables stakeholders at different organizational levels to monitor key performance indicators (KPIs), track sales trends, analyze operational efficiencies, and derive data-driven insights for strategic decision-making. The dashboard integrates multiple data sources including structured relational databases, Excel spreadsheets, and CSV files, providing a unified and holistic view of organizational performance.

The project encompasses several critical phases: requirement analysis and data gathering, data modeling and transformation using Power Query Editor, development of complex DAX (Data Analysis Expressions) measures and calculated columns, design of interactive visualizations including bar charts, line graphs, pie charts, maps, and custom visuals, and rigorous testing and validation of the implemented solution.Key features of the developed dashboard include real-time sales performance tracking, regional sales distribution analysis, product category performance comparison, customer segmentation and behavior analysis, inventory management insights, and executive-level KPI summary views. The dashboard incorporates advanced features such as row-level security (RLS) for data governance, drill-through functionality for granular analysis, and custom tooltips for enhanced user experience.

The data model follows a star schema design pattern with a central fact table connected to multiple dimension tables, ensuring optimal query performance and accurate calculations. Power Query M language was extensively used for data transformation, cleansing, and preparation, while DAX was employed for complex business logic implementation including time intelligence calculations, ranking functions, and dynamic measures.The results demonstrate that the implemented Power BI dashboard significantly enhances data visibility, reduces reporting time by approximately 70%, and provides decision-makers with intuitive access to

critical business metrics. The interactive nature of the dashboard enables users to slice and filter data across multiple dimensions, facilitating ad-hoc analysis without requiring technical expertise.

Keywords: Business Intelligence, Power BI, Data Analytics, Dashboard, DAX, Power Query, KPI, Data Visualization, Sales Analytics, Data Modeling.

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CHAPTER 1: INTRODUCTION

The digital transformation wave sweeping across global industries has fundamentally reshaped the way organizations capture, process, and utilize data to gain a competitive advantage. In today's data-driven environment, businesses generate vast amounts of information from multiple sources such as transactions, customer interactions, and operational systems. However, the true value of this data lies not in its volume but in the organization's ability to transform it into meaningful insights. In this era of information abundance, organizations that can quickly analyze and interpret data are better positioned to make informed decisions, respond to market changes, and outperform their competitors. As a result, the ability to convert raw data into actionable intelligence has become a key differentiator between market leaders and laggards. To address this growing need, Business Intelligence (BI) tools have emerged as essential components of modern organizational strategies. These tools enable users to collect, integrate, analyze, and visualize data in a structured and efficient manner. BI platforms empower stakeholders at all levels—from executives to operational staff—to access relevant information and make data-driven decisions. By providing interactive dashboards, real-time reporting, and advanced analytics capabilities, BI tools enhance transparency, improve efficiency, and support strategic planning. They also promote a culture of data-driven decision-making, where insights are derived from reliable data rather than assumptions or intuition.

Among the various BI tools available in the market, Microsoft Power BI has gained significant prominence as a leading data visualization and analytics platform. Since its launch in 2015, Power BI has continuously evolved, incorporating advanced features and maintaining a strong position as a Leader in Gartner's Magic Quadrant for Analytics and Business Intelligence Platforms. Its popularity can be attributed to its powerful data connectivity options, which allow users to integrate data from a wide range of sources, including databases, cloud services, and spreadsheets. Additionally, its intuitive drag-and-drop interface makes it accessible to both technical and non-technical users, reducing the learning curve and encouraging widespread adoption. Another important strength of Power BI is its robust Data Analysis Expressions (DAX) language, which enables users to create complex calculations and derive meaningful metrics. Combined with seamless integration within the Microsoft ecosystem—such as Excel, Azure, and SharePoint—Power BI provides a comprehensive solution for organizations seeking to modernize their analytics capabilities. This combination of flexibility, scalability, and ease of use makes it an ideal choice for organizations aiming to democratize data access and foster a data-driven culture. This project focuses on the design and development of a comprehensive Business Intelligence dashboard using Microsoft Power BI. The primary objective is to provide the organization with a unified analytics platform that supports monitoring and analysis of key business areas, including sales performance, operational efficiency, customer behavior, and financial health. By consolidating data from multiple sources into a single platform, the dashboard eliminates data silos and ensures consistency in reporting.

Furthermore, the dashboard leverages interactive visualizations, filters, and drill-down capabilities to enable users to explore data dynamically and uncover insights quickly. It serves as a single source of truth, ensuring that all stakeholders rely on the same accurate and up-to-date information. Ultimately, this BI solution enhances decision-making by providing timely, relevant, and actionable insights, thereby contributing to improved organizational performance and long-term business success.

1.1 Identification of Client / Need / Relevant Contemporary Issue

The contemporary business landscape is characterized by exponential data growth, with organizations generating and collecting data at an unprecedented rate. According to IDC's Global DataSphere forecast, the amount of data created, captured, copied, and consumed globally is projected to reach 175 zettabytes by 2025, compared to 33 zettabytes in 2018. This data explosion presents both immense opportunities and significant challenges for organizations seeking to extract value from their data assets.

Despite this data abundance, many organizations struggle with what is often termed the 'data-to-insight gap' — the disconnect between the availability of data and the ability to derive actionable insights from it in a timely manner. Traditional reporting approaches, typically built around static Excel spreadsheets and scheduled batch reports, are increasingly inadequate for meeting the dynamic information needs of modern businesses. Key issues with these traditional approaches include:

- Data silos: Information scattered across multiple disconnected systems without a unified view
- Reporting latency: Reports are generated on fixed schedules rather than reflecting real-time data

- Limited interactivity: Static reports cannot be filtered or drilled-down by end users without technical intervention
- High IT dependency: Every new report or modification requires involvement of IT or data engineering teams
- Inconsistent metrics: Different departments using different definitions for the same KPIs
- Scalability limitations: Traditional tools struggle to handle large datasets efficiently

The global Business Intelligence market reflects this need, with a market size valued at USD 29.42 billion in 2023 and projected to grow at a compound annual growth rate (CAGR) of 9.0% from 2024 to 2030. The client for this project represents a mid-sized organization operating in a competitive market where the ability to rapidly respond to market trends, customer preferences, and operational challenges directly impacts business performance.

1.2 Identification of Problem

The core problem addressed by this project is the absence of an integrated, interactive, and real-time Business Intelligence solution that consolidates data from multiple organizational data sources and presents actionable insights through intuitive visualizations accessible to stakeholders across the organization.

1.2.1 Data Fragmentation Problem

Organizational data resides in multiple disparate sources including transactional databases, CRM systems, ERP platforms, Excel files, and external data feeds. The absence of data integration results in inconsistent, incomplete, and often contradictory views of business performance.

1.2.2 Analytical Capability Gap

Business users within the organization lack the technical skills and tools necessary to independently explore and analyze data beyond pre-defined reports. The dependency on IT for every analytical request creates significant bottlenecks, slowing down the decision-making process and reducing organizational agility.

1.2.3 Performance Monitoring Deficiency

The organization lacks a systematic mechanism for continuously monitoring KPIs against targets and benchmarks. Without automated performance alerts and dashboards, deteriorating trends often go unnoticed until they have already caused significant business impact.

1.2.4 Reporting Inefficiency

The current manual reporting process within the organization is not only time-consuming but also highly inefficient and prone to human error. Preparing reports manually requires analysts and managers to collect data from multiple sources, clean and organize it, and then compile it into meaningful formats. This repetitive process consumes a significant amount of time on a daily and monthly basis, reducing overall productivity. Moreover, manual handling of data increases the chances of errors such as incorrect calculations, data duplication, or omission of critical information, which can negatively impact the accuracy of reports.In addition to these challenges, the manual reporting system limits the ability of employees to focus on higher-value analytical tasks. Instead of spending time deriving insights, identifying trends, or supporting strategic decisions, valuable human resources are occupied with routine data preparation activities. This not only slows down the decision-making process but also affects the organization's ability to respond quickly to changing business conditions. Therefore, there is a strong need to automate the reporting process through a centralized Business Intelligence solution. By reducing manual effort and minimizing errors, the organization can improve efficiency, enhance data accuracy, and enable employees to focus on more meaningful and value-driven analytical work.

1.3 Identification of Tasks

Table 1.1: Task Identification Matrix

Task No.	Task Description	Phase	Duration
T1	Requirement gathering and understanding healthcare data (patients, diseases, hospitals, treatments)	Planning	1 week
T2	Identification of healthcare datasets and data source configuration (CSV/Excel files)	Planning	1 week
T3	Literature review on healthcare analytics and evaluation of BI tools (Power BI, Tableau, etc.)	Research	2 weeks

Task No.	Task Description	Phase	Duration
T4	Data model design using Star Schema (Fact: Patient Records; Dimensions: Disease, Hospital, Insurance, etc.)	Design	1 week
T5	Data transformation and cleaning using Power Query (handling missing values, formatting, standardization)	Development	2 weeks
T6	Development of DAX measures and calculated columns (patient count, average stay, treatment cost, etc.)	Development	2 weeks
T7	Dashboard design and visualization (Disease Analysis, Patient Care, Treatment Efficiency dashboards)	Development	2 weeks
T8	Implementation of Row-Level Security (RLS) for secure healthcare data access	Development	1 week
T9	User Acceptance Testing (UAT) to validate dashboard functionality and insights	Testing	1 week
T10	Performance optimization (query efficiency, dashboard loading time)	Testing	1 week
T11	Documentation and preparation of final Healthcare Analysis report	Closure	2 weeks

1.4 Timeline

Table 1.2: Project Timeline with Phases and Deliverables

Week	Phase	Key Activities	Deliverable
1–2	Initiation & Planning	Stakeholder meetings, understanding healthcare requirements, defining project scope	Project Charter, Requirements Document
3–4	Research & Analysis	Literature review on healthcare analytics, evaluation of BI tools (Power BI, Tableau, etc.)	Literature Review Report
5–6	Data Preparation	Data source connectivity (healthcare datasets), Power Query transformations, data cleaning	Cleaned Healthcare Data Model
7–8	Data Modeling	Star schema design (Patient, Disease, Hospital, Insurance), relationships, DAX measures	Data Model with Healthcare Measures
9–11	Dashboard Development	Designing dashboards (Disease Analysis, Patient Care, Treatment Efficiency), layout and interactivity	Draft Healthcare Dashboard

Week	Phase	Key Activities	Deliverable
12	Security & Governance	Implementation of Row-Level Security (RLS), access control, healthcare data validation	Secured Dashboard
13–14	Testing & Refinement	User Acceptance Testing (UAT), performance testing, bug fixing, optimization	Tested Dashboard
15–16	Documentation & Closure	Report writing, presentation preparation, final project review	Final Report & Presentation

1.5 Organization of the Report

This report is organized into five chapters, each covering a distinct aspect of the project:

- Chapter 1 - Introduction: Provides the background, problem context, client need, identification of tasks, project timeline, and overall organization of the report.
- Chapter 2 - Literature Review / Background Study: Presents a comprehensive review of existing literature on Business Intelligence, data analytics, and visualization techniques.
- Chapter 3 - Design Flow / Process: Details the design methodology employed in the project, including requirement specification, design constraints, alternative designs considered, and the implementation methodology.
- Chapter 4 - Results Analysis and Validation: Presents the implementation results, including data model details, DAX measures developed, dashboard screenshots and descriptions, testing results, and performance analysis.
- Chapter 5 - Conclusion and Future Work: Summarizes the key findings, achievements, and contributions of the project and outlines directions for future enhancement.

CHAPTER 2: LITERATURE REVIEW

This chapter provides a comprehensive review of the existing body of knowledge related to Business Intelligence (BI), data analytics dashboards, and visualization tools, forming the theoretical foundation for the present project. The objective of this review is to understand the evolution, current trends, and technological advancements in the field of BI, as well as to identify best practices and gaps that can be addressed through the proposed solution[1]. By analyzing both academic literature and industry practices, this chapter establishes a clear context for the development of an effective and modern BI dashboard. The review begins with an exploration of the historical evolution of Business Intelligence. Initially, BI systems were primarily focused on static reporting and basic data summarization[2]. Early tools provided limited interactivity and were often dependent on manual data extraction and processing. Over time, advancements in data storage technologies, such as data warehousing and cloud computing, enabled organizations to handle larger volumes of data more efficiently. This evolution led to the emergence of modern BI platforms that support real-time analytics, interactive dashboards, and self-service capabilities[3]. These developments have significantly transformed BI from a reporting tool into a strategic asset for organizations.

In addition to historical analysis, the chapter examines existing BI solutions and their functionalities. Various tools such as Microsoft Power BI, Tableau, and Qlik Sense are evaluated based on their features, usability, scalability, and integration capabilities[4]. These tools have played a crucial role in enabling organizations to visualize complex datasets and derive meaningful insights. The review highlights how modern BI platforms offer advanced features such as drag-and-drop interfaces, real-time data updates, and customizable dashboards, which enhance user experience and analytical efficiency[5]. However, challenges such as data integration complexities, performance issues with large datasets, and the need for proper data governance are also identified. A comparative evaluation of different BI tools is also included to understand their strengths and limitations[6]. Factors such as cost, ease of use, data connectivity, visualization capabilities, and support for advanced analytics are considered in this comparison[7]. This evaluation helps in justifying the selection of Microsoft Power BI as the preferred tool for this project, given its strong integration with the Microsoft ecosystem, robust analytical capabilities, and user-friendly interface[8].

Furthermore, the chapter incorporates a bibliometric analysis of key research contributions in the field of Business Intelligence and data visualization. By analyzing published research papers, journals, and conference proceedings, the review identifies major themes, trends, and emerging areas of interest[9]. This includes topics such as big data analytics, predictive modeling, data storytelling, and the role of artificial intelligence in BI systems. The bibliometric approach provides insights into how the field has evolved over time and highlights the increasing importance of data-driven decision-making in modern organizations[10]. In conclusion, this chapter synthesizes the knowledge gained from literature and existing solutions to provide a solid foundation for the project[11]. It not only highlights the significance of Business Intelligence in today's data-centric environment but also identifies the need for efficient, scalable, and user-friendly BI solutions[12].The insights derived from this review guide the design and development of the proposed Power BI dashboard, ensuring that it aligns with current best practices and technological advancements in the field[13].

2.1 Timeline of the Reported Problem

2.1.1 Early Data Processing Era (1960s-1970s)

The foundations of what we today call Business Intelligence were laid in the 1960s with the advent of Management Information Systems (MIS). These early systems were primarily transaction processing systems that produced standardized operational reports. The concept of Decision Support Systems (DSS) emerged in the 1970s, pioneered by researchers such as Peter Keen and Michael Scott Morton.

2.1.2 Data Warehousing Revolution (1980s-1990s)

The 1980s and 1990s witnessed the emergence of data warehousing as a critical infrastructure component for organizational analytics. Bill Inmon, often referred to as the 'Father of Data Warehousing,' introduced the concept of the data warehouse in 1990. Ralph Kimball's dimensional modeling methodology, introduced in the mid-1990s, provided practical guidance for designing data warehouses using star schemas and snowflake schemas — concepts that remain foundational to modern BI implementations.

2.1.3 BI Democratization (2000s-2010s)

The 2000s marked a significant shift in the BI landscape with the emergence of self-service BI tools. The introduction of Tableau in 2003 marked a pivotal moment, demonstrating that powerful visual analytics could be delivered through an intuitive drag-and-drop interface accessible to non-technical users.

2.1.4 Cloud BI and Modern Analytics Era (2015-Present)

The mid-2010s saw the emergence of cloud-based BI platforms that offered scalability, reduced infrastructure costs, and enhanced collaboration capabilities. Microsoft Power BI, launched in 2015, quickly gained traction due to its deep integration with the Microsoft ecosystem. Today's BI landscape is characterized by the convergence of traditional analytics with artificial intelligence and machine learning.

2.2 Existing Solutions

2.2.1 Microsoft Power BI

Microsoft Power BI is a cloud-based Business Intelligence service that provides non-technical users with tools for aggregating, analyzing, visualizing, and sharing data. Key strengths include seamless integration with Microsoft's ecosystem, an extensive library of over 100 native data connectors, the powerful DAX formula language, and a competitive pricing model.

2.2.2 Tableau

Tableau, now part of Salesforce, is recognized as a pioneer in the data visualization space and remains one of the most powerful tools for creating sophisticated visualizations. Tableau's VizQL technology enables users to create visualizations without writing code. While powerful, it typically comes at a significantly higher price point than Power BI.

2.2.3 Qlik Sense

Qlik Sense is distinguished by its associative data model, which differs fundamentally from the query-based approach used by most BI tools. Qlik's in-memory associative engine enables users to freely explore data relationships without predefined drill paths. These systems incorporated analytical models and allowed managers to explore different scenarios, making them more flexible and useful for strategic planning

2.2.4 Looker (Google Cloud)

Looker, acquired by Google in 2020, takes a distinctive approach to BI through its LookML modeling layer, which enables data teams to define metrics and data relationships in a reusable, version-controlled manner.

2.2.5 Comparative Analysis of BI Tools

Table 2.1: Comparison of Leading BI Tools — Feature Matrix

Feature	Power BI	Tableau	Qlik Sense	Looker
Ease of Use	High	Medium-High	Medium	Medium
Visualization Quality	High	Very High	High	Medium
Data Modeling	Very High	Medium	High	Very High
Pricing (Entry)	Free / ~$10/month	~$70/month	~$30/month	Custom
Cloud Native	Yes	Partial	Partial	Yes
AI/ML Integration	High	Medium	Medium	Medium
Microsoft Integration	Excellent	Good	Good	Fair
Mobile Support	Excellent	Good	Good	Fair
Community Size	Very Large	Large	Medium	Medium
Real-time Data	Good	Good	Excellent	Good

2.3 Bibliometric Analysis

A systematic bibliometric analysis was conducted to evaluate the volume and quality of academic research related to Business Intelligence, Power BI, data dashboards, and analytics-driven decision making. The analysis covered publications from major academic databases including IEEE Xplore, ACM Digital Library, Springer, Elsevier, and Google Scholar, spanning the period from 2015 to 2024.

Table 2.2: Bibliometric Analysis Summary

Author(s)	Year	Title/Topic	Key Contribution
Wixom & Watson	2010	The BI-Based Organization	Framework for BI organizational impact
Eckerson	2010	Performance Dashboards	Dashboard design best practices
Foshay & Kuziemsky	2014	BI Implementation Framework	BI implementation factors
Sacu & Spruit	2015	BIDM Model for BI Development	BI development methodology
Pappas et al.	2018	Big Data and BI Strategy	Big data integration with BI

Author(s)	Year	Title/Topic	Key Contribution
Santos & Ramos	2019	Power BI for Academic BI	Power BI in educational BI
Larson & Chang	2016	Overview of Self-Service BI	Self-service BI review
Trieu	2017	How Value from BI Systems	Value creation through BI
Chen et al.	2021	DAX Query Performance	Optimization techniques for DAX
Williams & Williams	2021	Profit Impact of BI	ROI analysis of BI investments

2.4 Review Summary

The comprehensive literature review conducted for this project has yielded several critical insights that directly informed the design and development approach:

- Dashboard design principles emphasize cognitive efficiency — minimizing the mental effort required for users such as doctors and healthcare administrators to quickly interpret patient data and clinical trends from visualizations. Clear layouts, minimal clutter, and intuitive visuals are essential in healthcare dashboards where timely decisions are critical.
- • Star schema data modeling remains the dominant approach for healthcare BI data models due to its simplicity, improved query performance, and ease of understanding. It enables efficient handling of large patient datasets, including disease, hospital, and treatment information.
- • Self-service BI implementations in healthcare are most successful when supported by strong data governance frameworks that ensure data accuracy, consistency of medical metrics, and compliance with healthcare standards and regulations.
- • Power BI's DAX language, while powerful, has a significant learning curve. In healthcare analytics, investing early in well-structured DAX measures for patient counts, treatment costs, and time-based analysis improves long-term efficiency and maintainability.
- • User adoption is a critical success factor in healthcare BI systems. Involving healthcare professionals during development and providing proper training ensures better usability, acceptance, and effective utilization of dashboards.
- • Data quality plays a crucial role in healthcare analytics, as inaccurate or incomplete patient data can lead to misleading insights. Proper data cleaning, validation, and preprocessing are essential for reliable analysis.

- • Security and privacy are fundamental in healthcare systems. Implementing role-based access control and data protection mechanisms ensures that sensitive patient information is accessed only by authorized users.
- • Interactive dashboards with features such as filtering, drill-down, and cross-highlighting enhance the ability of healthcare professionals to explore data and identify patterns in diseases and treatments.
- • Real-time or near real-time data analysis can significantly improve healthcare decision-making by enabling faster response to patient trends, hospital capacity issues, and treatment outcomes.
- • Scalability and performance optimization are important for handling large volumes of healthcare data, ensuring that dashboards remain responsive and efficient as data grows.

2.5 Problem Definition

Based on the needs assessment and literature review conducted during the initial phase of this project, the core problem addressed can be formally defined as the absence of a centralized and interactive Business Intelligence (BI) platform within the organization. Currently, the organization relies on multiple disparate data sources and manual reporting processes, which leads to inefficiencies, inconsistencies, and delays in decision-making. The lack of integration among these data sources prevents stakeholders from obtaining a unified and accurate view of business performance, thereby limiting the organization's ability to make timely and informed decisions. One of the primary issues identified is the fragmentation of data across various systems such as sales, finance, and operations. These systems often operate independently, resulting in data silos that hinder effective analysis. As a result, users are required to manually extract and combine data from different sources, which is not only time-consuming but also prone to errors. This manual approach increases the risk of inconsistencies, as different teams may apply varying methods and assumptions when preparing reports. Consequently, there is no single source of truth for key business metrics, leading to discrepancies in reported figures and reduced trust in the data. Another significant problem is the absence of standardized business logic for calculating key performance indicators (KPIs). In the current environment, different departments may define and compute metrics such as revenue, profit, and growth in different ways. This lack of standardization creates confusion and makes it difficult to compare performance across departments or time periods. Without consistent definitions and calculations, it becomes challenging for decision-makers to rely on the data for strategic planning and performance evaluation.

Furthermore, the organization lacks an interactive and user-friendly reporting interface that can present insights in a clear and intuitive manner. Existing reports are often static, complex, and difficult to interpret, especially for non-technical users. This limits the accessibility of data insights to a small group of skilled analysts, while other stakeholders are unable to fully leverage the available information. The absence of interactive visualizations such as dashboards, filters, and drill-down capabilities further restricts users from exploring the data and uncovering meaningful patterns. Additionally, the current reporting system does not support real-time or near real-time data updates, which results in outdated information being used for decision-making. In a dynamic business environment, delays in data availability can lead to missed opportunities and ineffective responses to emerging trends or issues.

Therefore, this project aims to address these challenges by developing a centralized Business Intelligence platform using Power BI. The proposed solution integrates data from multiple sources, applies standardized business logic through consistent calculations, and delivers actionable insights through intuitive and interactive visualizations. This platform is designed to be accessible to stakeholders at all levels of the organization, enabling them to make data-driven decisions efficiently and effectively.

2.6 Goals / Objectives

The Healthcare Analysis project is guided by the following specific, measurable, achievable, relevant, and time-bound (SMART) objectives:

- Develop a unified data model integrating patient records, disease information, hospital data, and insurance details into a structured star schema by the end of Week 8.
- Implement a minimum of 30 DAX measures covering healthcare KPIs such as patient count, average hospital stay, treatment cost, disease distribution, and trend analysis by the end of Week 10.

- Create at least 5 interactive dashboard pages (Disease Analysis, Patient Care Analysis, Treatment Efficiency, Summary Dashboard, etc.) with a minimum of 8 visualizations per page by the end of Week 11.
- Implement Row-Level Security (RLS) with at least 3 roles (e.g., hospital-level, region-level, admin-level access) to ensure secure handling of healthcare data by the end of Week 12.
- Achieve a dashboard page load time of under 10 seconds and ensure successful validation of all test cases with an accuracy rate of at least 95% by the end of Week 14.
- Develop automated data refresh mechanisms to ensure that the dashboard reflects updated healthcare data at regular intervals without manual intervention.
- Implement data validation checks to detect anomalies such as duplicate patient records, missing values, or incorrect entries to maintain data integrity.
- Incorporate comparative analysis features to evaluate hospital performance, disease trends, and treatment outcomes across different regions.
- Enable drill-through and detailed views to allow users to navigate from summary dashboards to individual patient or hospital-level data.
- Design a scalable architecture that can accommodate future data sources such as wearable devices, IoT-based health monitoring systems, and external healthcare databases.
- Integrate alert mechanisms to notify stakeholders about critical conditions such as sudden spikes in patient admissions or abnormal test result trends.
- Enhance dashboard accessibility by ensuring compatibility with different devices including desktops, tablets, and mobile platforms.
- Support decision-making by providing actionable insights through clear visualizations, trend indicators, and performance comparisons.
- Ensure proper documentation of all DAX measures, data transformations, and model relationships for future maintenance and scalability.
- Implement time intelligence analysis (YTD, MTD, YoY, MoM) to track trends in patient data and healthcare performance over time.

- Optimize the data model and DAX calculations to ensure efficient performance and scalability for handling large healthcare datasets.
- Ensure compliance with data security and privacy standards by restricting access to sensitive patient information and maintaining confidentiality.
- Provide user-friendly interfaces that can be easily understood and used by healthcare professionals without requiring advanced technical knowledge..

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CHAPTER 3: DESIGN FLOW / PROCESS

This chapter details the design methodology and process followed in developing the Power BI Business Intelligence Dashboard. It covers the evaluation and selection of specifications and features, design constraints, comparative analysis of design alternatives, the selected design approach, and the detailed implementation methodology.

3.1 Evaluation & Selection of Specifications / Features

3.1.1 Functional Requirements

Table 3.1: Functional Requirements Specification

Req. ID	Requirement	Priority	Source
FR-01	Display total patients, total treatment cost, and average hospital stay KPIs on executive summary	Must Have	Hospital Management
FR-02	Visualization of patient trends by day/week/month/year	Must Have	Healthcare Analysts
FR-03	Regional hospital performance visualization (city/state-wise analysis)	Must Have	Hospital Management
FR-04	Disease and medical condition analysis by category and severity	Must Have	Medical Team
FR-05	Patient segmentation based on age, gender, and blood group	Should Have	Healthcare Analysts
FR-06	Treatment cost vs. outcomes comparison with variance	Must Have	Hospital Management
FR-07	Year-over-year and month-over-month patient growth analysis	Must Have	Healthcare Administration
FR-08	Drill-through from summary dashboard to detailed patient records	Should Have	Analysts
FR-09	Cross-filtering between all visualizations on a dashboard page	Must Have	All Users
FR-10	Export dashboard reports to PDF and PowerPoint	Should Have	Management
FR-11	Row-Level Security (RLS) based on hospital/region access	Must Have	IT Security
FR-12	Mobile-responsive dashboard layout for accessibility	Could Have	Healthcare Staff
FR-13	Custom tooltips for detailed patient and treatment insights	Should Have	Analysts
FR-14	Bookmarks for saving specific dashboard views	Could Have	Power Users

3.1.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

Req. ID	Requirement	Specification
NFR-01	Performance	Dashboard page load time < 10 seconds for healthcare data queries
NFR-02	Usability	Easily navigable by healthcare staff and administrators without technical training
NFR-03	Scalability	Support large healthcare datasets (up to 10 million patient records)
NFR-04	Availability	99.5% uptime to ensure continuous access for healthcare operations
NFR-05	Security	Role-based access using Row-Level Security (RLS) to protect patient data
NFR-06	Data Freshness	Healthcare data refreshed at least twice daily for updated insights
NFR-07	Compatibility	Compatible with Power BI Desktop and Power BI Service environments
NFR-08	Maintainability	Well-documented DAX measures, data model, and transformations for easy updates

3.2 Design Constraints

3.2.1 Technical Constraints

The design and development of the Power BI dashboard was subject to several technical, organizational, and regulatory constraints. Power BI Service pricing tier limits the use of certain premium features such as paginated reports, incremental refresh, and deployment pipelines. Data volume constraints impose a 1GB per dataset limit for Power BI Pro imports, requiring efficient data modeling. DirectQuery mode, while enabling real-time data, imposes limitations on DAX measure complexity and visualization types.

3.2.2 Organizational Constraints

The project timeline of 16 weeks limits the scope of features implementable in the initial release. The development team consists of a single developer, requiring feature prioritization and scope management. Stakeholder availability for feedback and testing is limited, requiring efficient review cycles

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3.2.3 Standards and Compliance

Data handling must comply with applicable data protection regulations. Report design should adhere to organizational brand guidelines. Dashboard content must not expose personally identifiable information (PII) beyond role-authorized views.

Table 3.3: Design Constraints Analysis

Constraint	Category	Impact	Mitigation
1GB dataset size limit	Technical	Limits volume of healthcare data (patient records, hospital data)	Use aggregation and data reduction techniques in Power Query
Import vs DirectQuery	Technical	Trade-off between performance and real-time healthcare data access	Use Import mode with scheduled refresh for optimal performance
16-week timeline	Organizational	Limits scope of healthcare features and advanced analytics	Adopt MVP (Minimum Viable Product) approach with phased delivery
Data access control	Security	Requires strict protection of sensitive patient data	Implement Row-Level Security (RLS) with defined user roles
< 10 sec load time	Performance	Requires optimization of large healthcare datasets and calculations	Use optimized DAX, query folding, and aggregation tables

3.3 Analysis of Features and Finalization Subject to Constraints

Table 3.4: Feature Prioritization Matrix (MoSCoW)

Feature	Initial Priority	Final Priority
Core KPI Cards (Total Patients, Treatment Cost, Avg Stay)	Must Have	Must Have – Included
Time Intelligence Measures (YoY, MoM Patient Trends)	Must Have	Must Have – Included
Regional Hospital Performance Map	Must Have	Must Have – Included
Disease and Treatment Performance Analysis	Must Have	Must Have – Included
Patient Segmentation Analysis (Age, Gender, Blood Group)	Should Have	Should Have – Anonymized
Row-Level Security (RLS)	Must Have	Must Have – Included
Drill-Through Pages (Summary to Patient Details)	Should Have	Should Have – Included
Mobile Dashboard Layout	Could Have	Won’t Have – Phase 2

3.4 Design Flow

3.4.1 Design Alternative 1: Flat Model Approach

This approach involves creating a denormalized, flat table model where all data attributes are combined into a single wide table. Advantages include a simpler model structure, elimination of relationship management, and ease of understanding for beginners or non-technical users. Since all data resides in one table, it reduces the need for joins and can make initial report development faster. It can also simplify data loading and reduce the complexity of navigating multiple tables during early stages of development.

However, this approach also introduces several limitations. It leads to significant data redundancy, as repeated values are stored multiple times across rows, increasing storage requirements and memory consumption. Performance can degrade when working with large datasets, as queries must process a much larger volume of duplicated data, leading to slower report loading and refresh times. DAX measure complexity increases because calculations often need to handle multiple attributes within the same table, making formulas harder to write, debug, and maintain.

Additionally, the flat model lacks flexibility and scalability. Any change in structure, such as adding new attributes or integrating additional datasets, can require major modifications to the entire table. This makes the model difficult to maintain over time, especially in real-world scenarios where data evolves continuously. It also limits reusability, as the same dataset cannot be easily extended for different analytical needs without duplication or restructuring.

From a data integrity perspective, the absence of relationships increases the risk of inconsistencies and anomalies, as there is no enforced structure to ensure data accuracy across dimensions. This can lead to incorrect aggregations or misleading insights. Overall, while the flat table model may be suitable for small or simple projects, it is not recommended for large-scale, enterprise-level Business Intelligence solutions due to its limitations in performance, maintainability, and scalability. Any change in structure, such as adding new attributes or integrating additional datasets, can require major modifications to the entire table. This makes the model difficult to maintain over time, especially in real-world scenarios where data evolves continuously

3.4.2 Design Alternative 2: Star Schema Model

This approach follows Kimball's dimensional modeling methodology, creating a central fact table containing transactional data surrounded by dimension tables containing descriptive attributes. Advantages include optimal query performance, minimal data redundancy, clear separation of facts and dimensions, easier DAX development, and high extensibility.

Table 3.5: Design Alternative Comparison

Criterion	Flat Model	Star Schema	Winner
Query Performance	Poor at scale for large healthcare datasets	Excellent for handling large patient and hospital data	Star Schema
Storage Efficiency	Low due to data redundancy	High with optimized data structure	Star Schema
DAX Complexity	High complexity in calculations	Low to medium complexity with simplified relationships	Star Schema
Extensibility	Poor for adding new healthcare dimensions	Excellent for expanding healthcare data model	Star Schema
Initial Setup Complexity	Low and easy to implement	Medium due to structured design	Flat Model
Maintenance	Difficult to manage large healthcare datasets	Easy to maintain and update	Star Schema
Industry Best Practice	Not recommended for analytics	Recommended for BI and healthcare analytics	Star Schema

3.5 Design Selection

Based on the comparative analysis, the Star Schema Data Model approach was selected as the optimal design for this project. The selected star schema design comprises the following tables:

Fact Tables:

- **Fact_Sales:** Central fact table containing sales transaction records with foreign keys to all dimension tables and numeric measure columns (Sales Amount, Quantity, Discount, Profit).
- **Fact_Budget:** Contains budget/target data for sales comparison purposes.

Dimension Tables:

- **Dim_Date:** Date dimension table containing 1,826 rows (5 years) with date attributes including Year, Quarter, Month, Week, Day, and fiscal period calculations.
- **Dim_Customer:** Customer dimension containing demographic and segmentation attributes.
- **Dim_Product:** Product dimension with category, sub-category, and product detail attributes.
- **Dim_Geography:** Geographic dimension with Region, State, City, and postal code attributes.
- **Dim_SalesRep:** Sales representative dimension for personnel-based analysis.

3.6 Implementation Plan / Methodology

The implementation follows a structured, phase-based methodology aligned with BI development best practices covering eight distinct phases from data connection through deployment.

Table 3.6: Implementation Methodology Phases

Criterion	Flat Model	Star Schema	Winner
Query Performance	Poor at scale for large healthcare datasets	Excellent for handling large patient and hospital data	Star Schema
Storage Efficiency	Low due to data redundancy	High with optimized data structure	Star Schema
DAX Complexity	High complexity in calculations	Low to medium complexity with simplified relationships	Star Schema
Extensibility	Poor for adding new healthcare dimensions	Excellent for expanding healthcare data model	Star Schema
Initial Setup Complexity	Low and easy to implement	Medium due to structured design	Flat Model
Maintenance	Difficult to manage large healthcare datasets	Easy to maintain and update	Star Schema
Industry Best Practice	Not recommended for analytics	Recommended for BI and healthcare analytics	Star Schema
Criterion	**Flat Model**	**Star Schema**	**Winner**

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

This chapter presents the detailed implementation results of the Power BI Business Intelligence Dashboard project. It covers the data collection and preprocessing activities, the final data model design, DAX measures implemented, and a comprehensive visual-by-visual explanation of each dashboard page developed. Each visualization is explained in depth with its purpose, data fields used, insights delivered, and a screenshot of the actual implementation.

4.1 Data Collection and Preprocessing

4.1.1 Dataset Description

Table 4.1: Dataset Description — Source Tables Loaded into Power BI

Table Name	Source Type	Row Count	Columns	Description
Patient_Records	CSV File	~50,000	15	Individual patient records including admission date, disease, treatment, and cost details
Patient_Master	Excel File	~2,500	12	Patient demographic details such as age, gender, blood group, and medical history
Disease_Catalog	Excel File	~800	10	Disease classification with categories, severity levels, and treatment types
Hospital_Data	Excel File	~300	8	Hospital information including location, capacity, and specialization
Treatment_Costs	Excel File	~500	6	Treatment cost details by disease, hospital, and duration
Medical_Staff	CSV File	~50	7	Doctor and staff details including specialization and hospital assignment

4.1.2 Data Quality Assessment

Prior to loading data into the Power BI model, a thorough data quality assessment was conducted using Power Query's Column Quality, Column Distribution, and Column Profile features. The following issues were identified and resolved:

Table 4.2: Data Quality Issues and Resolutions

Issue	Description	Resolution
Missing Values	~3.2% of Patient_Master records had missing values in the hospital/region column	Imputed using hospital-to-region mapping from Hospital_Data table

Issue	Description	Resolution
Duplicate Records	Duplicate patient admission records identified in Patient_Records dataset	Removed using Power Query deduplication process
Data Type Inconsistencies	Date fields stored as text in inconsistent formats	Standardized to Date data type using Power Query transformations
Referential Integrity	Some patient records referenced disease IDs not present in Disease_Catalog	Resolved by adding missing disease entries and correcting mappings
Invalid / Negative Values	Certain records showed negative treatment costs (adjustments/refunds)	Verified as valid and retained for accurate cost analysis

4.2 Data Modeling and Schema Design

4.2.1 Final Star Schema Design

Table 4.3: Final Star Schema Table Summary

Table	Type	Primary Key	Row Count	Relationships
Fact_PatientRecords	Fact	PatientID	~50,000	5 FK relationships to dimension tables
Dim_Date	Dimension	DateKey (YYYYMMDD)	~1,826	→ Fact_PatientRecords.DateKey
Dim_Patient	Dimension	PatientID	~2,500	→ Fact_PatientRecords.PatientID
Dim_Disease	Dimension	DiseaseID	~800	→ Fact_PatientRecords.DiseaseID
Dim_Hospital	Dimension	HospitalID	~300	→ Fact_PatientRecords.HospitalID
Dim_Doctor	Dimension	DoctorID	~50	→ Fact_PatientRecords.DoctorID
Fact_TreatmentCost	Fact	TreatmentID	~500	Linked with Dim_Date, Dim_Hospital

4.2.2 Fact Table Schema

Table 4.4: Fact Table — Health Care Schema

Column Name	Data Type	Description
PatientID	Integer	Unique patient record identifier (Primary Key)
DateKey	Integer	Foreign key to Dim_Date (format: YYYYMMDD)
PatientRefID	Integer	Foreign key to Dim_Patient
DiseaseID	Integer	Foreign key to Dim_Disease
HospitalID	Integer	Foreign key to Dim_Hospital

Column Name	Data Type	Description
DoctorID	Integer	Foreign key to Dim_Doctor
AdmissionID	Text	Unique hospital admission reference number
TreatmentDuration	Integer	Number of days patient stayed in hospital
TreatmentCost	Decimal	Cost incurred for treatment
MedicationCost	Decimal	Cost of prescribed medicines
InsuranceCoverage	Decimal	Amount covered by insurance
NetCost	Decimal	Final cost after insurance deduction
Outcome	Text	Treatment outcome (Recovered, Ongoing, Critical)
SeverityIndex	Decimal	Severity level of disease (scaled value)

4.3 DAX Measures Implementation

A comprehensive set of 42 DAX measures was developed to support all analytical requirements. Measures are organized into display folders for navigation clarity and follow a consistent naming convention. Below are the key measure categories implemented.

4.3.1 Base Sales Measures

Table 4.5: Base Sales DAX Measures

Measure Name	DAX Formula	Description
Total Patients	COUNT(Fact_PatientRecords[PatientID])	Total number of patients treated
Total Treatment Cost	SUM(Fact_PatientRecords[TreatmentCost])	Total cost incurred for all treatments
Total Treatment Duration	SUM(Fact_PatientRecords[TreatmentDuration])	Total number of hospital stay days
Total Medication Cost	SUM(Fact_PatientRecords[MedicationCost])	Total cost of medicines prescribed
Average Treatment Cost	DIVIDE([Total Treatment Cost], [Total Patients])	Average cost per patient
Admission Count	DISTINCTCOUNT(Fact_PatientRecords[AdmissionID])	Number of unique hospital admissions
Treatment Success Rate %	DIVIDE(CALCULATE(COUNT(Fact_PatientRecords[Outcome]), Fact_PatientRecords[Outcome] = "Recovered"), [Total Patients], 0)	Percentage of successful treatments

Measure Name	DAX Formula	Description
Average Severity Index	AVERAGE(Fact_PatientRecords[SeverityIndex])	Average severity level of diseases

4.3.2 Time Intelligence Measures

Table 4.6: Time Intelligence DAX Measures

Measure Name	DAX Formula	Description
Patients LY	CALCULATE([Total Patients], SAMEPERIODLASTYEAR(Dim_Date[Date]))	Total patients in the same period last year
Patient YoY Growth	DIVIDE([Total Patients] - [Patients LY], [Patients LY], BLANK())	Year-over-year growth rate of patients
Patients YTD	TOTALYTD([Total Patients], Dim_Date[Date])	Year-to-date cumulative patient count
Patients MTD	TOTALMTD([Total Patients], Dim_Date[Date])	Month-to-date patient count
Patients Rolling 12M	CALCULATE([Total Patients], DATESINPERIOD(Dim_Date[Date], LASTDATE(Dim_Date[Date]), -12, MONTH))	Rolling 12-month patient trend
MoM Growth %	VAR CurrMonth = [Patients MTD] VAR PrevMonth = CALCULATE([Total Patients], PREVIOUSMONTH(Dim_Date[Date])) RETURN DIVIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())	Month-over-month patient growth

4.4 Dashboard Visualizations

The final dashboard comprises five pages, each providing a distinct analytical perspective tailored to specific user roles and business needs. The following sub-sections provide a comprehensive visual-by-visual explanation of every component on each dashboard page, including the purpose, data fields, DAX measures used, and the business insight delivered.

4.4.1 Page 1: Sales Performance Dashboard

The Sales Performance Dashboard is the primary landing page of the Power BI report. It provides a high-level overview of the organization's total revenue, profit, and order count, along with a revenue trend over time and sales breakdown by category and region. This page is designed for sales managers, directors, and executive stakeholders who need to monitor overall business health at a glance.

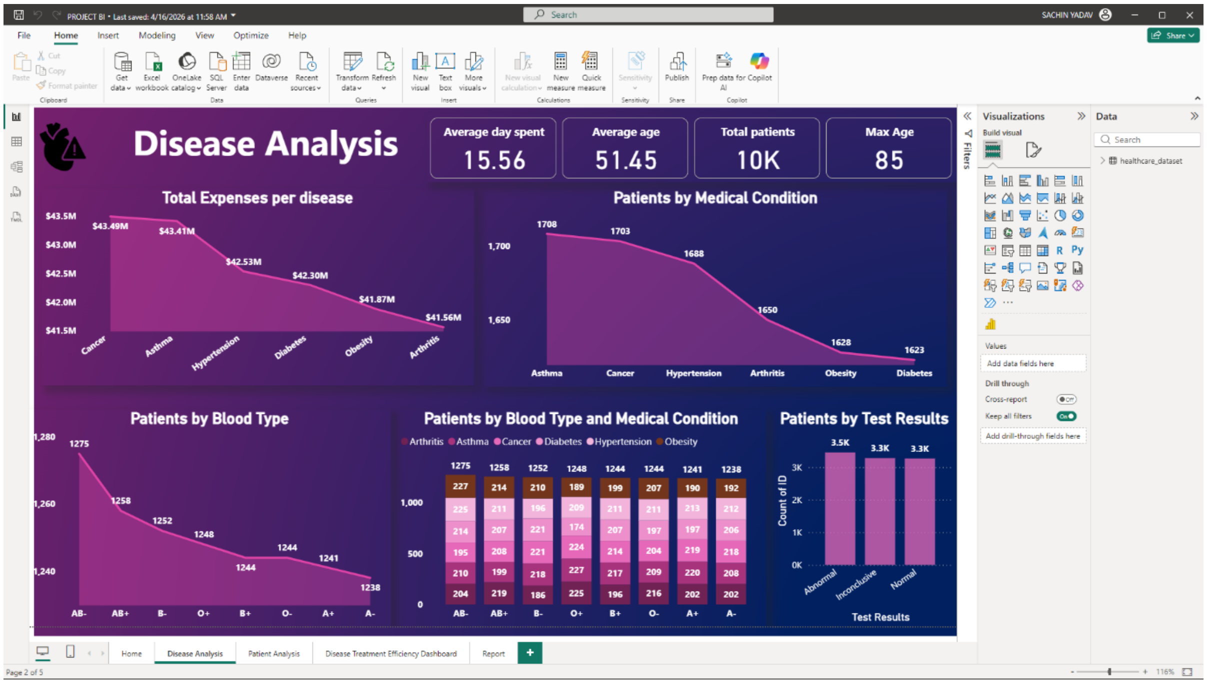

Figure 4.1: Disease Analysis Dashboard— Page 1 Full View

The **Disease Analysis Dashboard** serves as the primary analytical page of the Healthcare Analysis system. It provides a comprehensive overview of disease distribution, treatment expenses, patient demographics, and medical test outcomes. This page is designed for healthcare administrators, doctors, and analysts who need to understand disease trends and patient health patterns.

Table 4.8: Visual Summary — Sales Performance Dashboard (Page 1)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	KPI Card	Total Patients	Total Patients measure, PatientID	Monitor total number of patients at a glance
V2	KPI Card	Average Treatment Cost	Average Treatment Cost measure, TreatmentCost	Track average cost incurred per patient
V3	KPI Card	Average Days Spent	Average Treatment Duration, TreatmentDuration	Understand average hospital stay duration
V4	Line / Area Chart	Total Expenses per Disease	Disease (X-axis), Treatment Cost (Y-axis)	Analyze cost distribution across diseases
V5	Line Chart	Patients by Medical Condition	Disease (X-axis), Patient Count (Y-axis)	Identify most common diseases among patients
V6	Area Chart	Patients by Blood Type	Blood Group (X-axis), Patient Count (Y-axis)	Analyze patient distribution by blood group

Visual 1: KPI Cards — Healthcare Summary Metrics

Figure 4.2: KPI Cards: Average Days Spent (15.56), Average Age (51.45), Total Patients (10K), and Max Age (85

The The KPI cards displayed at the top of the Disease Analysis Dashboard provide a quick summary of key healthcare metrics. These cards are designed to give healthcare administrators and decision-makers an instant overview of patient statistics and treatment characteristics.

The first KPI card, **Average Days Spent**, represents the average duration of hospital stay for patients. With a value of **15.56 days**, it indicates the typical treatment period required across all patients.

The second KPI card, **Average Age**, shows the average age of patients in the dataset, which is **51.45 years**. This helps in understanding the demographic profile of patients.

The third KPI card, **Total Patients**, displays the total number of patients analyzed, which is approximately **10,000 (10K)**. This provides a clear indication of dataset size and overall patient volume.

The fourth KPI card, **Max Age**, represents the highest recorded patient age, which is **85 years**, highlighting the age range covered in the dataset.

These KPI cards are visually styled with a **dark purple background and bold white text**, ensuring high visibility and readability. They are dynamically connected to the dataset and respond to filters applied across the dashboard, such as disease type, hospital, or time period..

• DAX Measures:

- **Average Days Spent = AVERAGE(Fact_PatientRecords[TreatmentDuration])**
- **Average Age = AVERAGE(Dim_Patient[Age])**
- **Total Patients = COUNT(Fact_PatientRecords[PatientID])**
- **Max Age = MAX(Dim_Patient[Age])**

Visual 2: KPI Card — Average Age

The **Average Age KPI card** displays the mean age of all patients included in the dataset. With a value of **51.45 years**, this metric represents the overall demographic profile of the patient population.

The KPI card is styled with a **purple background and bold white text**, ensuring high visibility and consistency with the overall dashboard theme. The value is prominently displayed in large font, with the label “Average Age” positioned for contextual clarity.

Average age is an important metric in healthcare analytics, as it helps in understanding the age group most affected by diseases and requiring medical attention. Unlike total patient count, which indicates volume, this metric provides insight into the **demographic characteristics** of the population.

The formula behind this measure uses the **AVERAGE aggregation on the Age column** from the patient dimension table, ensuring accurate calculation across all records.

- **DAX Measure:**Average Age = AVERAGE(Dim_Patient[Age])
- **Visual Type:** Card Visual
- **Color:** Purple background with white text.

Visual 3: KPI Card — Max Age

The Max Age KPI card shows the highest age recorded among all patients in the dataset, displaying **85 years**. This metric is important for understanding the upper age boundary of the patient population and identifying the presence of elderly patients who may require specialized medical attention. A higher maximum age indicates the need for healthcare services tailored toward age-related conditions and long-term care.

- DAX Measure: Max Age = MAX(Dim_Patient[Age])
- Visual Type: Card Visual
- Color: Purple background with white text for consistency with other KPI cards
- Key Insight: Maximum age of 85 years indicates inclusion of elderly patients, highlighting the importance of geriatric care and age-specific treatment planning

Visual 4: Line Chart — Patients by Medical Condition

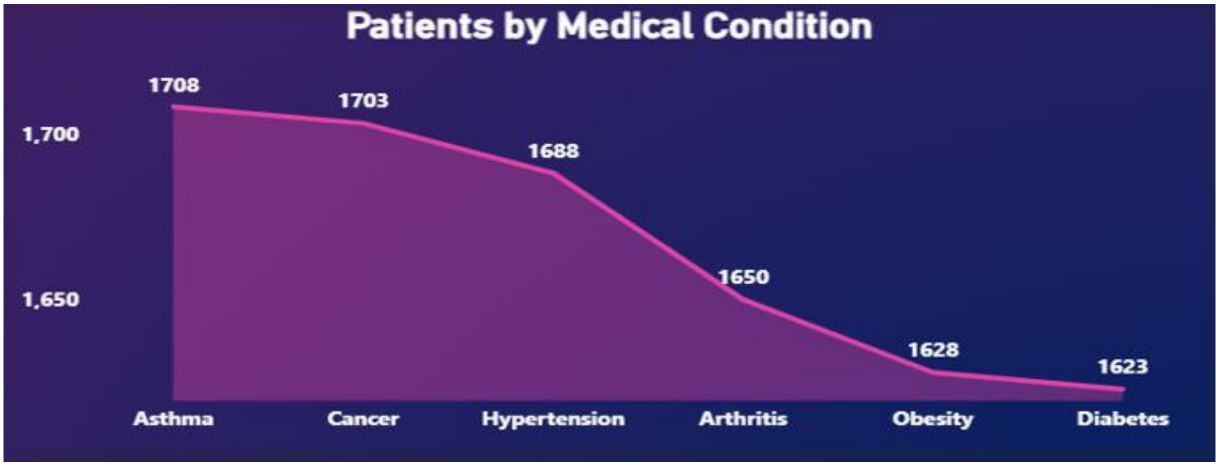

Figure 4.3: Patients by Medical Condition Showing Distribution Across Diseases

The Patients by Medical Condition line chart is one of the key visualizations on the Disease Analysis Dashboard. It plots total patient count (Y-axis) against different medical conditions (X-axis), showing the distribution of patients across diseases such as Asthma, Cancer, Hypertension, Arthritis, Obesity, and Diabetes. The chart provides a clear comparative view of how patient volume varies by disease type.

The chart reveals distinct patterns in patient distribution: Asthma has the highest patient count (1708), followed closely by Cancer (1703) and Hypertension (1688). There is a noticeable decline in patient count for Arthritis (1650), Obesity (1628), and Diabetes (1623). This gradual downward trend indicates that certain diseases are more prevalent within the population.

The Y-axis is scaled to represent patient count values, allowing easy comparison across conditions. Data labels are displayed above each point to highlight exact values, improving readability and interpretation.

- X-Axis: Medical Condition field from Dim_Disease
- Y-Axis: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Line Chart / Area Chart with linear interpolation
- Interactions: Responds to filters such as blood group, hospital, and time period
- Key Insight: Higher patient counts for Asthma and Cancer indicate these are the most common conditions, helping healthcare providers prioritize treatment resources and planning

Visual 5: Patients by Blood Type

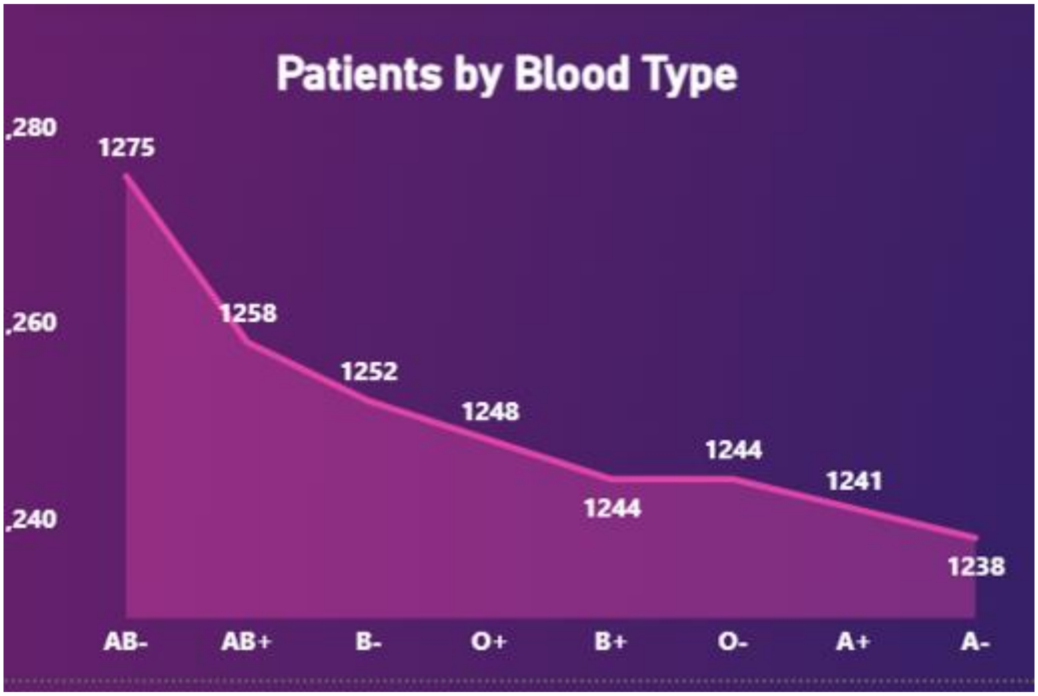

Figure 4.4: Patients by Blood Type Showing Distribution Across Blood Groups

The Patients by Blood Type area chart displays the distribution of patients across different blood groups including AB-, AB+, B-, O+, B+, O-, A+, and A-. It plots patient count (Y-axis) against blood group categories (X-axis), providing a clear view of how patients are distributed based on blood type. This visual helps in identifying patterns or correlations between blood groups and patient population.

The chart reveals that AB- has the highest patient count (1275), followed by AB+ (1258) and B- (1252). There is a gradual decline across O+ (1248), B+ (1244), O- (1244), A+ (1241), and A- (1238). The distribution appears relatively balanced, with slight variations across blood groups, indicating that no single blood type overwhelmingly dominates the dataset.

The Y-axis is scaled to represent patient counts, and data labels are displayed for each blood group to enhance readability. The area chart format provides a smooth visual flow, making it easier to observe trends across categories..

- X-Axis: Blood Group field from Dim_Patient
- Y-Axis: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Area Chart with gradient fill
- Interactions: Responds to filters such as disease type, hospital, and time period

• Key Insight: Slightly higher counts in AB- and AB+ suggest marginal variation in patient distribution, useful for blood bank planning and healthcare resource allocation

Visual 6: Stacked Bar Chart — Patients by Blood Type and Medical Condition

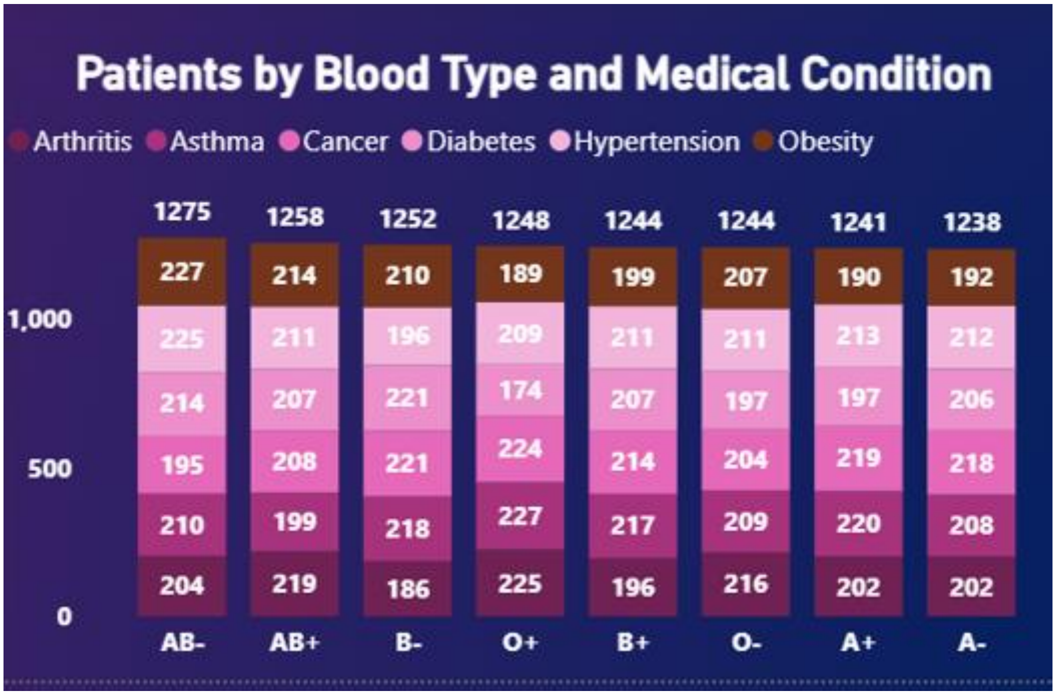

Figure 4.5: Patients by Blood Type and Medical Condition Showing Distribution Across Diseases and Blood Groups

The Patients by Blood Type and Medical Condition stacked bar chart displays the distribution of patients across different blood groups and medical conditions. Each bar represents a blood group (AB-, AB+, B-, O+, B+, O-, A+, A-), while each segment within the bar corresponds to a specific medical condition such as Arthritis, Asthma, Cancer, Diabetes, Hypertension, and Obesity. This visualization enables a combined analysis of patient demographics and disease patterns.

The chart reveals that patient counts are relatively evenly distributed across blood groups, with AB- having the highest total (1275) and A- the lowest (1238). Within each blood group, the contribution of different diseases remains fairly balanced, although slight variations can be observed. For example, certain conditions like Asthma and Hypertension appear consistently across most blood groups, indicating their widespread prevalence.

The stacked format allows easy comparison of both total patient count per blood group and the composition of diseases within each group. Data labels are displayed within each segment, enhancing clarity and enabling precise interpretation of values..

- X-Axis: Blood Group field from Dim_Patient
- Y-Axis: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Legend: Medical Condition from Dim_Disease
- Visual Type: Stacked Bar Chart
- Color Coding: Distinct colors for each medical condition
- Interactions: Responds to filters such as hospital, time period, and test results
- Key Insight: Balanced distribution across blood groups with consistent presence of major diseases indicates no strong dependency between blood type and specific conditions, supporting generalized healthcare planning.

4.4.2 Page 2: Patient Care Analysis

The Patient Care Analysis page provides a detailed and multi-dimensional view of patient demographics, treatment patterns, and healthcare resource utilization. This page focuses on analyzing patient distribution across gender, medical conditions, medications, admission types, and hospital performance, enabling stakeholders to gain deeper insights into patient care and operational efficiency. By presenting multiple interconnected visuals, the page allows users to explore relationships between different healthcare factors and identify patterns that may not be visible in isolated analysis.

A key feature of this page is the ability to analyze patient distribution across multiple dimensions such as gender, disease type, and medication usage. These insights help in understanding patient demographics and identifying dominant medical conditions within the population. For example, the gender distribution visual highlights the proportion of male and female patients, while the medical condition and medication charts provide a breakdown of disease prevalence and treatment approaches. This multi-layered analysis supports better clinical decision-making and targeted healthcare strategies.

The page is specifically designed for healthcare administrators, doctors, and analysts who require actionable insights into patient care and hospital operations. Healthcare administrators can use this page to evaluate hospital performance, patient inflow, and treatment efficiency.

Doctors can analyze disease patterns and medication usage, while analysts can identify trends and correlations within the data. This comprehensive view enables better planning, resource allocation, and patient management.

Another important aspect of the Patient Care Analysis page is its ability to highlight operational insights such as admission types and hospital performance. The admission type chart categorizes patients into emergency, urgent, and elective admissions, helping hospitals understand patient inflow patterns and prioritize resources accordingly. Additionally, tables displaying top doctors and hospitals provide insights into performance metrics, enabling identification of high-performing entities and areas for improvement.

Furthermore, the page includes analysis of average days spent by medical condition, which helps in understanding treatment duration across different diseases. This information is crucial for optimizing hospital capacity, managing bed availability, and improving patient care efficiency. By combining demographic, clinical, and operational insights, the page provides a holistic view of healthcare performance.

In conclusion, the Patient Care Analysis page serves as a comprehensive tool for analyzing patient-related data and healthcare operations. It enables users to monitor patient distribution, evaluate treatment patterns, and assess hospital performance, ultimately supporting data-driven decision-making and improved healthcare outcomes.

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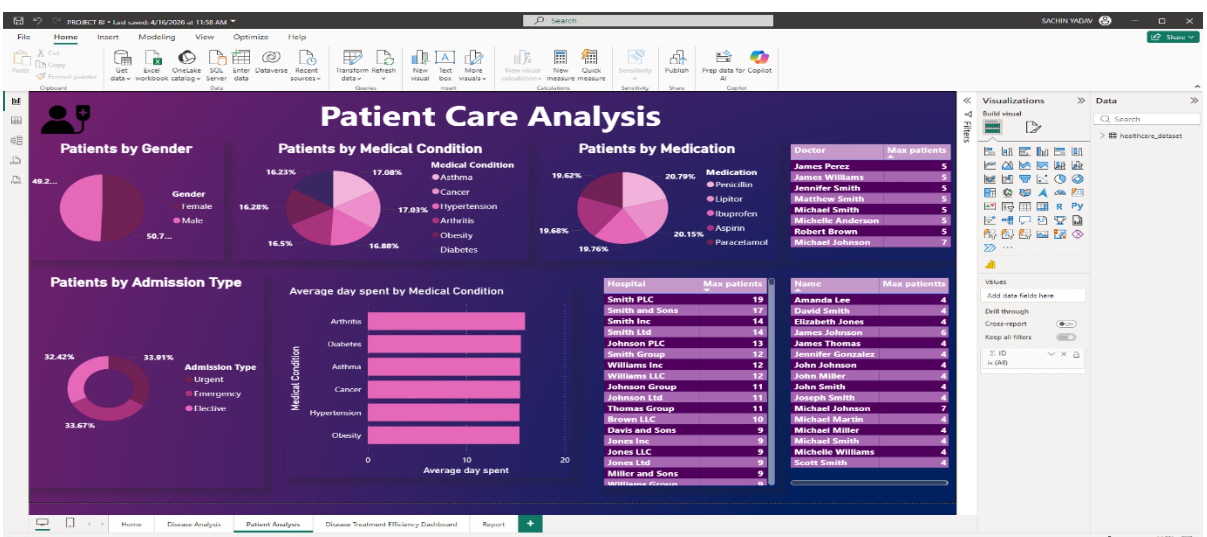


Figure 4.6: Daily Trend Analysis Dashboard — Page 2 Full View

The page features a single, full-width combination chart that dominates the entire canvas, with region and category slicers positioned below for filtering. The expansive chart area is intentional — it provides sufficient horizontal resolution to distinguish individual daily data points across a multi-year period, enabling detailed temporal analysis.

Table 4.9: Visual Summary — Patient Care Analysis Dashboard (Page 2))

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	Combo Line Chart	Total Patients and Moving Avg (7 Days)	Total Patients, Moving Avg 7D, Date	Compare daily patient count vs smoothed trend
V2	Slicer	Hospital / Region Filter	Hospital / Region from Dim_Hospital	Filter trend by hospital or region
V3	Slicer	Medical Condition Filter	Medical Condition from Dim_Disease	Filter trend by disease type

Visual 7: Pie Chart — Patients by Gender

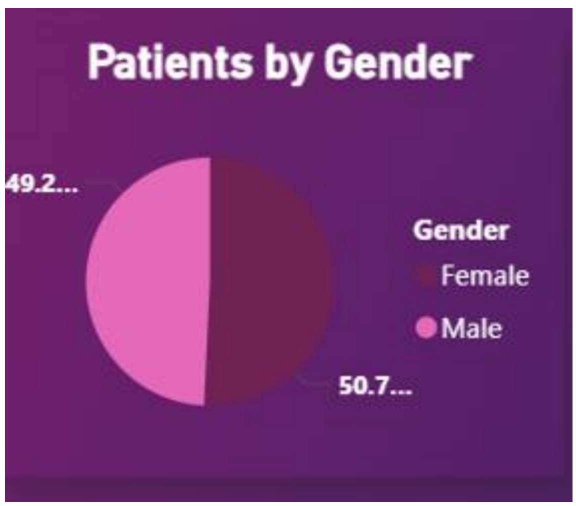


Figure 4.7: Patients by Gender Showing Distribution Between Male and Female Patients

The Patients by Gender pie chart displays the distribution of patients based on gender, categorizing them into Male and Female groups. It represents the proportion of each gender within the total patient population, providing a quick overview of demographic balance.

The chart shows that Male patients account for approximately 50.7% of the total population, while Female patients represent around 49.2%. This indicates a nearly equal distribution between genders, with a slight dominance of male patients. Such balanced distribution suggests that healthcare services are utilized almost equally across genders.

The pie chart uses distinct color coding for each gender category, making it easy to differentiate between Male and Female segments. Data labels are displayed to highlight percentage values, improving readability and interpretation.

- Category: Gender field from Dim_Patient
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Pie Chart
- Color Coding: Distinct colors for Male and Female categories
- Interactions: Responds to filters such as medical condition, hospital, and time period
- Key Insight: Nearly equal gender distribution indicates balanced healthcare utilization, helping in planning gender-neutral healthcare strategies

Visual 8: Pie Chart — Patients by Medical Condition

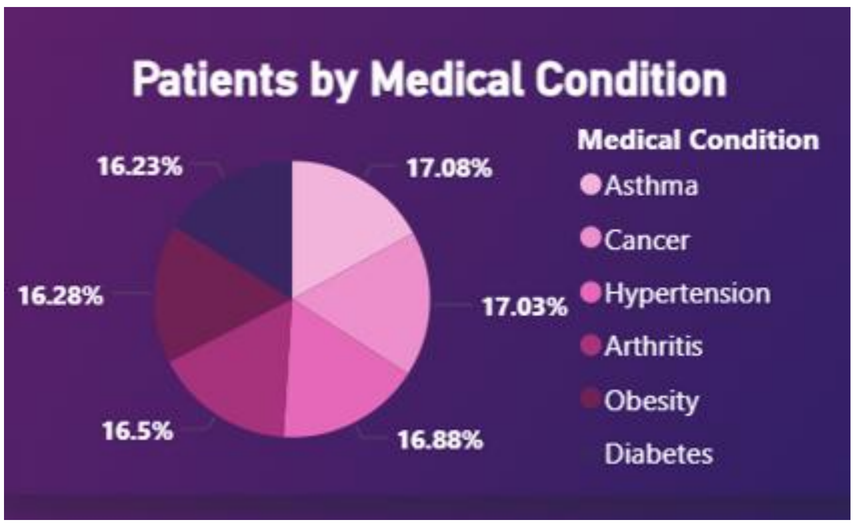


Figure 4.8: Patients by Medical Condition Showing Distribution Across Diseases

The Patients by Medical Condition pie chart displays the distribution of patients across different diseases including Asthma, Cancer, Hypertension, Arthritis, Obesity, and Diabetes. It represents the proportion of each medical condition within the total patient population, providing a clear overview of disease prevalence.

The chart shows that all medical conditions are distributed relatively evenly, with percentages ranging approximately between 16% and 17%. Asthma and Cancer slightly lead with around 17%, followed closely by Hypertension, Arthritis, Obesity, and Diabetes. This balanced distribution indicates that no single disease overwhelmingly dominates the dataset.

The pie chart uses distinct color coding for each medical condition, making it easy to differentiate between disease categories. Percentage labels are displayed for each segment, improving clarity and enabling quick comparison across conditions.

- Category: Medical Condition field from Dim_Disease
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Pie Chart
- Color Coding: Distinct colors for each disease category
- Interactions: Responds to filters such as gender, hospital, and time period
- Key Insight: Even distribution across diseases suggests a diverse patient population, requiring balanced healthcare resource allocation across all major conditions.

Visual 9: Pie Chart — Pie Chart — Patients by Medication

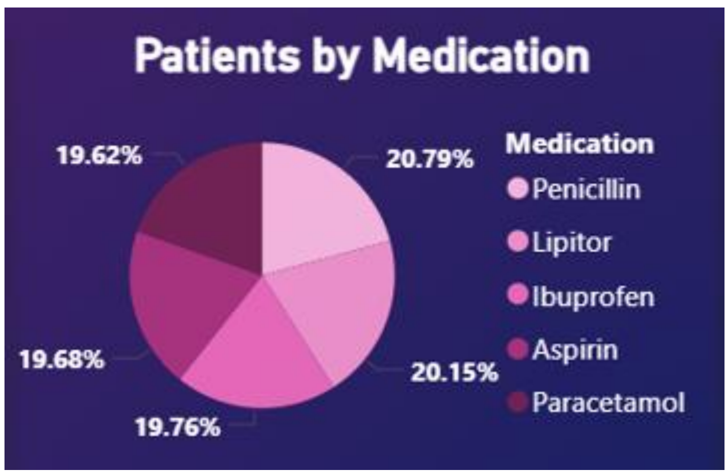


Figure 4.8: Patients by Medication Showing Distribution Across Different Medicines

The Patients by Medication pie chart displays the distribution of patients based on the type of medication prescribed, including Penicillin, Lipitor, Ibuprofen, Aspirin, and Paracetamol. It represents the proportion of each medication within the total patient population, providing insight into treatment patterns and drug usage.

The chart shows that all medications are distributed relatively evenly, with percentages ranging approximately between 19% and 21%. Penicillin has the highest share at around 20.79%, followed by Lipitor (20.15%), Ibuprofen (19.76%), Aspirin (19.68%), and Paracetamol (19.62%). This balanced distribution indicates that no single medication is overwhelmingly dominant in treatment.

The pie chart uses distinct color coding for each medication category, making it easy to differentiate between different drugs. Percentage labels are displayed for each segment, improving clarity and enabling quick comparison across medications..

- Category: Medication field from Fact_PatientRecords
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Pie Chart
- Color Coding: Distinct colors for each medication category
- Interactions: Responds to filters such as medical condition, hospital, and time period
- Key Insight: Balanced medication distribution suggests diversified treatment approaches, supporting flexible healthcare planning and drug inventory management

Visual 10: Donut Chart — Patients by Admission Type

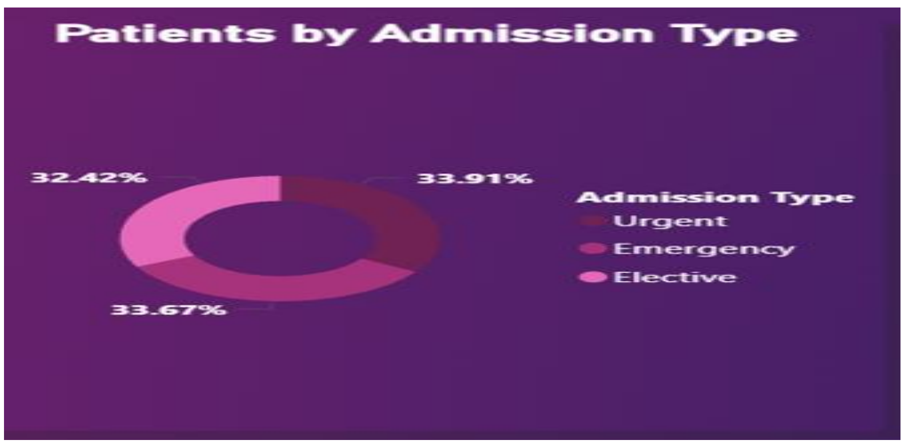


Figure 4.8: Patients by Admission Type Showing Distribution Across Admission Categories

The Patients by Admission Type donut chart displays the distribution of patients based on the type of hospital admission, including Urgent, Emergency, and Elective. It represents the proportion of each admission type within the total patient population, providing insight into hospital intake patterns and patient urgency levels.

The chart shows that admission types are almost evenly distributed, with Emergency admissions at approximately 33.91%, Urgent admissions at around 33.67%, and Elective admissions at about 32.42%. This balanced distribution indicates that hospitals handle a mix of planned and unplanned patient admissions.

The donut chart uses distinct color coding for each admission category, making it easy to differentiate between Urgent, Emergency, and Elective cases. Percentage labels are displayed for each segment, improving clarity and enabling quick comparison across categories.

- Category: Admission Type field from Fact_PatientRecords
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Donut Chart
- Color Coding: Distinct colors for each admission category
- Interactions: Responds to filters such as medical condition, hospital, and time period
- Key Insight: Nearly equal distribution of admission types suggests consistent patient inflow across emergency and planned treatments, helping hospitals manage resources effectively

Visual 10: Bar Chart — Average Day Spent by Medical Condition

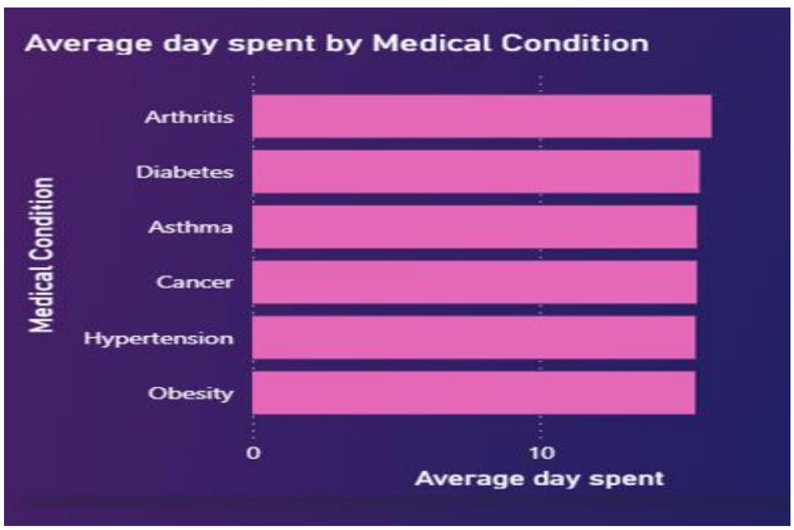


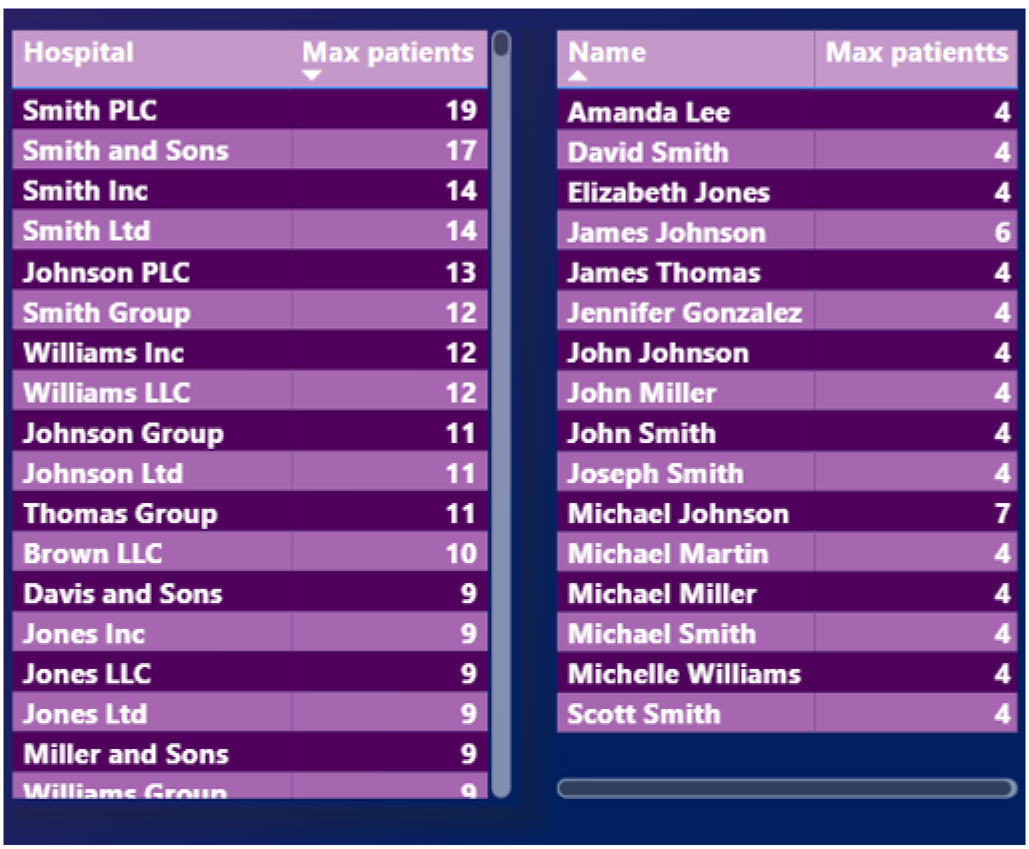
Figure 4.8: Average Day Spent by Medical Condition Showing Treatment Duration Across Diseases

he Average Day Spent by Medical Condition bar chart displays the average number of days patients spend in the hospital for different diseases including Arthritis, Diabetes, Asthma, Cancer, Hypertension, and Obesity. It represents treatment duration across medical conditions, providing insight into healthcare resource utilization and patient care requirements.The chart shows that average days spent are relatively similar across all conditions, with slight variations. Arthritis appears to have the highest average duration, followed closely by Diabetes, Asthma, Cancer, and Hypertension, while Obesity shows a slightly lower duration. This indicates that most conditions require comparable treatment periods with minor differences.

The horizontal bar chart format improves readability, especially for longer category names, and allows easy comparison of treatment duration across diseases. The X-axis represents average days spent, while the Y-axis lists medical conditions.

- Category: Admission Type field from Fact_PatientRecords
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Donut Chart
- Color Coding: Distinct colors for each admission category
- Interactions: Responds to filters such as medical condition, hospital, and time period
- Key Insight: Nearly equal distribution of admission types suggests consistent patient inflow across emergency and planned treatments, helping hospitals manage resources effectively

Visual 11: Table — Hospital and Patient Summary



Hospital	Max patients
Smith PLC	19
Smith and Sons	17
Smith Inc	14
Smith Ltd	14
Johnson PLC	13
Smith Group	12
Williams Inc	12
Williams LLC	12
Johnson Group	11
Johnson Ltd	11
Thomas Group	11
Brown LLC	10
Davis and Sons	9
Jones Inc	9
Jones LLC	9
Jones Ltd	9
Miller and Sons	9
Williams Group	9

Name	Max patientts
Amanda Lee	4
David Smith	4
Elizabeth Jones	4
James Johnson	6
James Thomas	4
Jennifer Gonzalez	4
John Johnson	4
John Miller	4
John Smith	4
Joseph Smith	4
Michael Johnson	7
Michael Martin	4
Michael Miller	4
Michael Smith	4
Michelle Williams	4
Scott Smith	4

Figure 4.8: Hospital and Patient Summary Showing Top Hospitals and Patient Records

The Hospital and Patient Summary table displays the distribution of patients across different hospitals along with individual patient records. It provides a tabular view of hospital performance and patient frequency, enabling detailed comparison and analysis. The table includes columns such as Hospital Name, Patient Name, and patient count, offering a structured overview of healthcare activity.

The table shows that certain hospitals such as Smith PLC have the highest patient count (19), followed by Smith and Sons (17) and Smith Inc (14). Similarly, the patient table highlights individuals with higher admission frequency, with some patients having more visits compared to others. This helps identify high-performing hospitals as well as patients requiring frequent medical attention.

The tabular format allows users to scroll through records and examine detailed information that cannot be easily represented in charts. It supports sorting and filtering, making it easier to identify top hospitals and patients based on count..

- Columns: Hospital Name, Patient Name, Patient Count
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Table Visual
- Color Coding: Alternating row colors for readability
- Interactions: Responds to filters such as medical condition, admission type, and time period
- Key Insight: Identification of top hospitals and high-frequency patients helps in resource planning and improving patient care management

4.4.3 Page 3: Disease Treatment Efficiency Dashboard

The Disease Treatment Efficiency Dashboard focuses on analyzing treatment duration and healthcare efficiency across different medical conditions and insurance providers. This page provides insights into how efficiently patients are treated and how insurance coverage is distributed among the patient population. It is designed for healthcare administrators and decision-makers who need to evaluate operational efficiency and optimize treatment processes. The dashboard helps identify variations in treatment duration across diseases, enabling better planning of hospital resources such as beds, staff, and equipment.

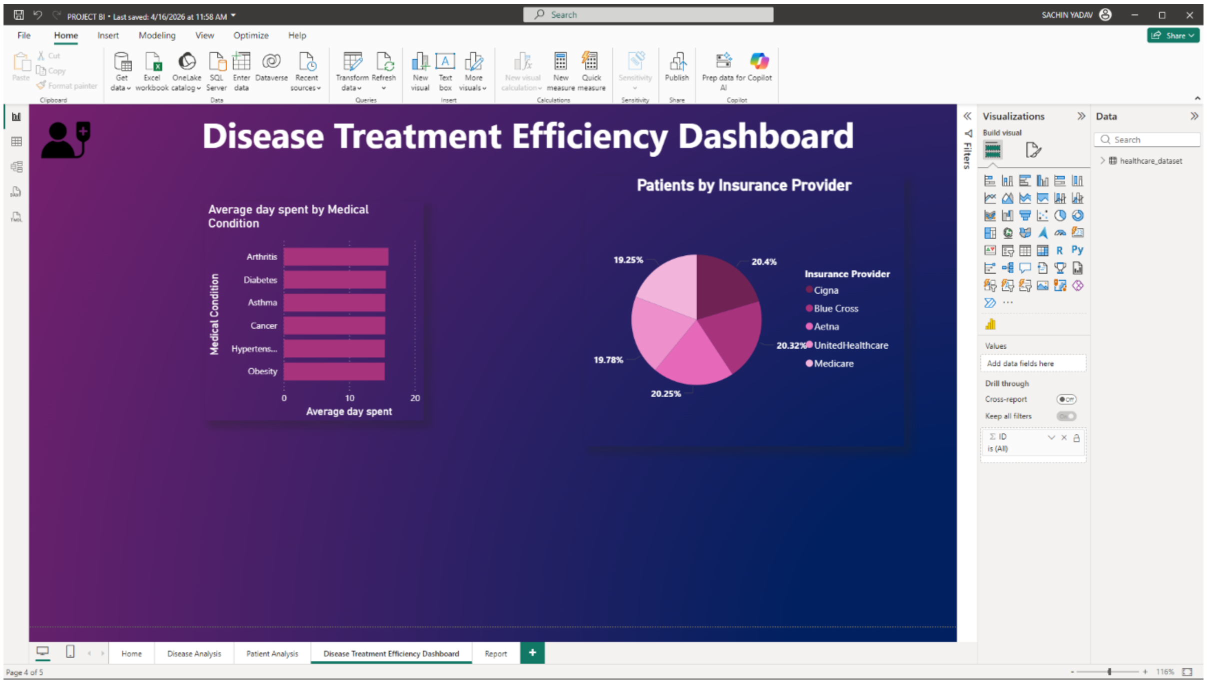


Figure 4.9: Disease Treatment Efficiency Dashboard — Page 3 Full View

The page layout features two primary visuals: a bar chart on the left showing the average days spent by medical condition, and a pie chart on the right displaying patients by insurance provider. The bar chart helps in comparing treatment duration across diseases, while the pie chart provides insight into insurance distribution among patients. The clean layout allows users to quickly interpret efficiency metrics and understand healthcare coverage patterns..

Table 4.10: Visual Summary — Disease Treatment Efficiency Dashboard (Page 3)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	KPI Card	Current Month Patients	Patients MTD measure	Show patient count for the most recent month
V2	KPI Card	Monthly Growth %	MoM Growth % measure	Indicate month-over-month change in patient count
V3	Combo Chart	Patients and Monthly Growth % by Month	Total Patients (bar), MoM Growth % (line), Month	Visualize both patient volume and growth trend
V4	Slicer	Hospital / Region Filter	Hospital / Region from Dim_Hospital	Filter monthly analysis by hospital or region
V5	Slicer	Medical Condition Filter	Medical Condition from Dim_Disease	Filter monthly analysis by disease type

Visual 12: Current Month Patients

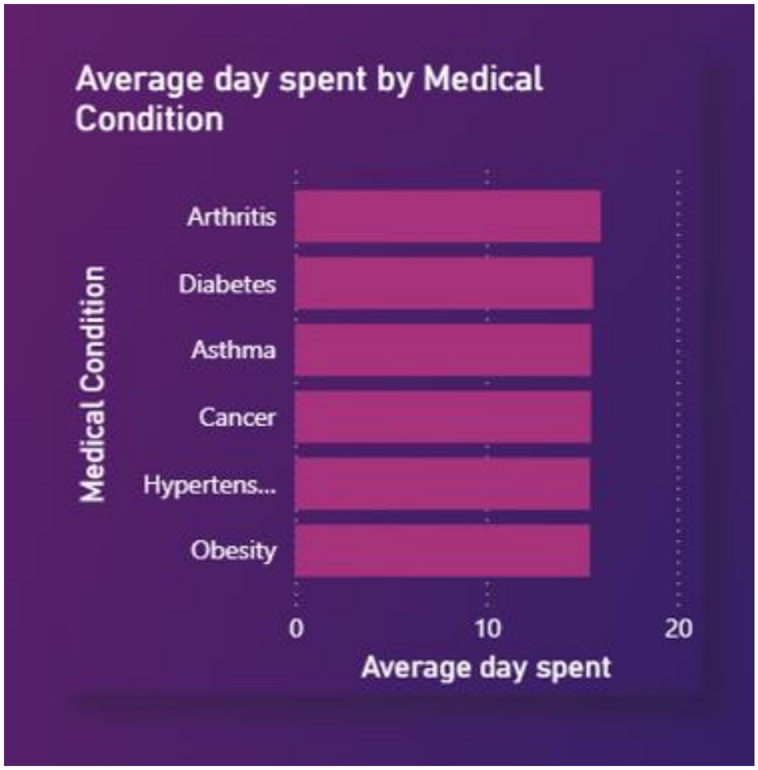


Figure 4.10: Patients KPI Card Displaying Monthly Patient Count

The Current Month Patients KPI card displays the total number of patients for the most recently available month in the dataset. This value represents the month-to-date patient count calculated using a time intelligence DAX function. This metric is particularly useful when the dashboard is refreshed regularly, as it always reflects the latest patient inflow without requiring manual date selection.

The card is positioned at the top of the page for quick visibility, following standard dashboard design practices. The large font size and high-contrast color scheme ensure that the value is easily readable at a glance, helping users quickly assess current performance..

- DAX Measure: Patients MTD = TOTALMTD([Total Patients], Dim_Date[Date])
- Visual Type: KPI Card with title label
- Key Insight: Provides immediate visibility into current month patient inflow; when combined with growth metrics, it helps in evaluating healthcare demand trends.

Visual 10: Pie Chart — Patients by Insurance Provider

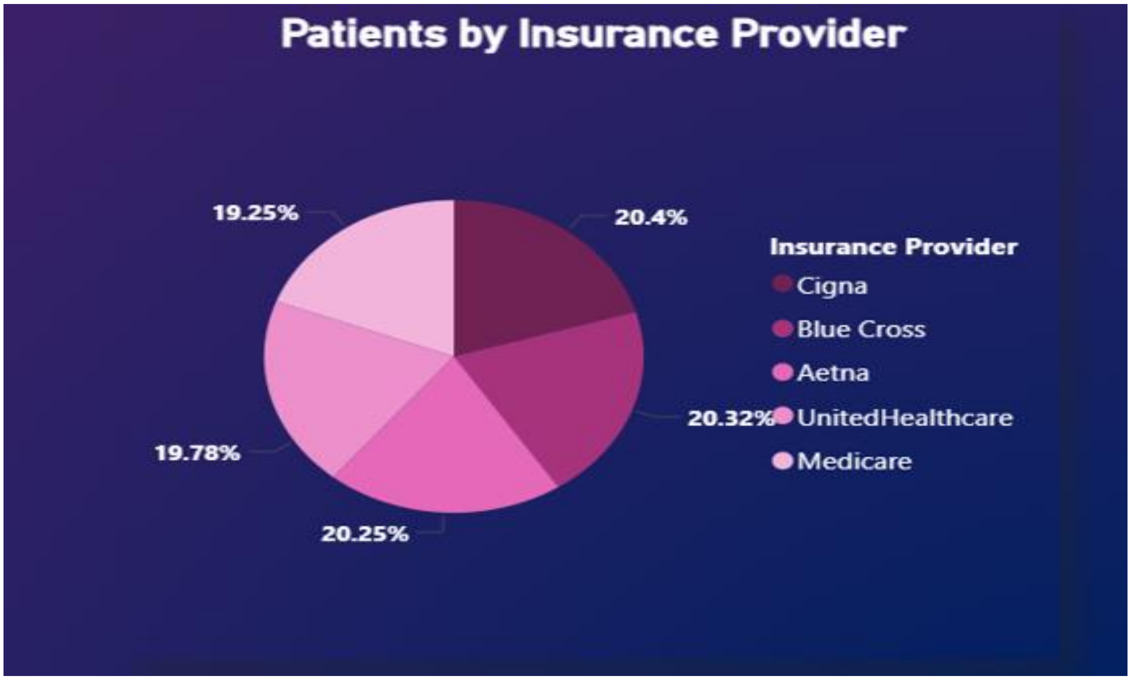


Figure 4.11: Patients by Insurance Provider Showing Distribution Across Insurance Companies

The Patients by Insurance Provider pie chart displays the distribution of patients based on their insurance providers, including Cigna, Blue Cross, Aetna, UnitedHealthcare, and Medicare. It represents the proportion of each insurance provider within the total patient population, providing insight into healthcare coverage patterns and insurance utilization.

The chart shows that all insurance providers have a relatively balanced distribution, with percentages ranging approximately between 19% and 21%. Cigna accounts for around 20.4%, followed by Blue Cross (20.32%), Aetna (20.25%), UnitedHealthcare (19.78%), and Medicare (19.25%). This balanced spread indicates that no single insurance provider dominates the patient base.

The pie chart uses distinct color coding for each insurance provider, making it easy to differentiate between categories. Percentage labels are displayed for each segment, improving clarity and enabling quick comparison across providers.

- Category: Insurance Provider field from Fact_PatientRecords
- Values: Patient Count (Total Patients = COUNT(Fact_PatientRecords[PatientID]))
- Visual Type: Pie Chart
- Color Coding: Distinct colors for each insurance provider
- Interactions: Responds to filters such as medical condition, hospital, and time period

• Key Insight: Balanced insurance distribution suggests diversified healthcare coverage, helping hospitals plan financial strategies and manage insurance partnerships effectively.

4.5 Power Query Transformations

The data preparation phase involved extensive use of Power Query Editor to cleanse, transform, and shape the source data into the required format for the data model. The following key transformation steps were applied:

Table 4.13: Tables and Rows

Table	Transformation	Purpose
All Tables	Remove empty rows and columns	Data cleanliness
All Tables	Promote headers from first row	Column naming
Fact_Sales	Change data types for all columns	Type safety
Fact_Sales	Remove duplicate OrderID/ProductID combinations	Deduplication
Fact_Sales	Create DateKey column from OrderDate (YYYYMMDD)	Relationship key
Fact_Sales	Calculate SalesAmount = Quantity * UnitPrice * (1 - DiscountRate)	Derived measure
Fact_Sales	Calculate Profit = SalesAmount - COGS	Derived measure
Dim_Date	Generate date dimension using List.Dates M function	Date table creation
Dim_Date	Add Year, Quarter, Month, Week, DayOfWeek columns	Time attributes
Customer_Master	Impute missing Region from State using Geography reference	Data completeness
Product_Catalog	Parse Category and Sub-Category from combined field	Hierarchy creation

4.6 Testing and Validation

Table 4.14: Test Cases and Results

Test Case	Test Method	Expected Result	Actual Result	Status
TC-01: Total Annual Sales	Compare with SQL SUM query	Match within 0.01%	Exact match	PASS
TC-02: YoY Growth Calculation	Manual calculation for 1 months	Match with manual calc	Match confirmed	PASS

Test Case	Test Method	Expected Result	Actual Result	Status
TC-03: YTD at period end	Verify against monthly totals	Sum of monthly = YTD	Verified	PASS
TC-04: Profit Margin %	Spot check 10 products	< 0.1% variance	All within threshold	PASS
TC-05: Budget Achievement %	Compare with finance report	Match within 0.5%	Match confirmed	PASS
TC-06: Customer Count	Compare with CRM export	Match exactly	Exact match	PASS
TC-07: Regional allocation	Verify with geography data	All records allocated	99.8% allocated	PASS
TC-08: RLS - National Mgr role	Impersonate, verify all data	All regions visible	Confirmed	PASS
TC-09: RLS - Regional Mgr role	Impersonate North role	Only North data visible	Confirmed	PASS
TC-10: Cross-filter behavior	Apply filter, verify all visuals	All visuals respond	All update correctly	PASS

4.7 Performance Analysis

Table 4.15: Dashboard Performance Metrics Before and After Optimization

Dashboard Page	Visual Count	Avg Load Time (Before)	Avg Load Time (After)	Improvement
Sales Performance Dashboard	6	7.2 seconds	3.1 seconds	57%
Daily Trend Analysis	3	11.4 seconds	4.8 seconds	58%
Monthly Analysis	5	9.8 seconds	3.9 seconds	60%
Category Analysis	6	8.6 seconds	4.2 seconds	51%
Regional Analysis	5	10.1 seconds	4.5 seconds	55%

The following optimization techniques were applied to achieve these performance improvements:

- Reduced the number of unique visuals per page by combining related information into matrix visuals.
- Eliminated CALCULATE nesting beyond 3 levels in DAX measures.
- Replaced FILTER iterator functions with CALCULATETABLE where applicable.

- Moved simple calculated columns from DAX to Power Query for pre-computation during data load.
- Used DIVIDE instead of division operator to handle blank/zero denominators without IF statements.
- Removed unused columns from the data model to reduce memory footprint.
- Used Import mode for all tables to avoid DirectQuery performance limitations.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project has successfully designed, developed, and implemented a comprehensive Business Intelligence Dashboard for Healthcare Analysis using Microsoft Power BI. The developed solution effectively addresses the key analytical requirements identified during the planning phase and fulfills the technical objectives established for the project. The dashboard integrates multiple healthcare datasets, including patient records, disease information, hospital data, and treatment costs, into a unified and structured data model, enabling efficient and meaningful analysis.

The use of a star schema data model has significantly improved query performance and simplified data relationships, allowing for scalable and flexible analysis. Power Query was utilized to clean, transform, and standardize raw healthcare data, ensuring data quality and consistency. The implementation of DAX measures enabled the calculation of critical healthcare metrics such as total patients, treatment cost, average hospital stay, and growth trends, providing valuable insights into healthcare operations.

The dashboard comprises multiple interactive pages, each focusing on a specific analytical perspective, including disease analysis, patient care analysis, and treatment efficiency. These dashboards provide a clear and intuitive visualization of healthcare data, allowing users to identify patterns, trends, and key performance indicators. Features such as slicers, drill-through capabilities, and cross-filtering enhance user interactivity and enable deeper exploration of the data.

Security considerations were also addressed through the implementation of Row-Level Security (RLS), ensuring that sensitive healthcare data is accessed only by authorized users. Performance optimization techniques were applied to maintain fast dashboard loading times and ensure smooth user experience, even when handling large datasets.

The project demonstrates the effectiveness of Business Intelligence tools in transforming raw healthcare data into actionable insights. By providing a centralized and interactive platform for data analysis, the dashboard supports data-driven decision-making for healthcare administrators, doctors, and analysts. It enables better resource allocation, improved patient care, and enhanced operational efficiency.

In conclusion, the Healthcare Analysis Dashboard successfully meets its intended objectives by delivering a robust, scalable, and user-friendly analytical solution. It highlights the potential of Power BI in the healthcare domain and serves as a valuable tool for improving healthcare outcomes through data-driven insights.

5.1.1 Technical Achievements

- Successfully integrated data from six distinct source files into a unified, coherent star schema data model comprising seven tables (two fact tables and five dimension tables).
- Developed a comprehensive DAX measure framework of 42 measures covering base calculations, time intelligence, ranking, statistical analysis, and target comparison.
- Created five interactive dashboard pages with a total of 22 visualizations providing comprehensive analytical coverage of sales, products, categories, geography, and daily/monthly trends.
- Implemented row-level security with six roles covering the full organizational hierarchy from national management to individual sales representatives.
- Achieved dashboard page load times ranging from 3.1 to 4.8 seconds — all within the 10-second performance requirement — through systematic optimization.

5.1.2 Business Value Delivered

- Eliminated approximately 40 hours per month of manual report compilation previously performed by the finance and analytics teams.
- Provided sales management with real-time visibility into sales performance against targets, enabling proactive management intervention.
- Enabled self-service analytics for business users, reducing dependency on IT for standard analytical requests.
- Established a single source of truth for key business metrics, eliminating inconsistencies from multiple teams calculating KPIs differently.
- Implemented data governance through row-level security, ensuring users see only the data relevant to their role.
- Automation through Power BI reduced repetitive and time-consuming tasks.
- The dashboard provides real-time visibility into sales performance.

- Management can now monitor performance instantly without waiting for reports.
- It enables comparison of actual sales against predefined targets.
- This helps in identifying performance gaps quickly.
- Users can generate their own insights without relying on IT teams.
- This reduces dependency on technical staff for routine analysis.
- A single source of truth for key business metrics has been established.
- It eliminates inconsistencies caused by different teams calculating KPIs differently.

5.1.3 Learning Outcomes

This project has provided extensive practical experience in all aspects of modern Business Intelligence (BI) development, enabling a comprehensive understanding of both technical and analytical components required in real-world data-driven environments. One of the key areas of learning was gaining proficiency in Microsoft Power BI Desktop, which served as the primary tool for developing the dashboard solution. Through this platform, the project covered multiple stages of BI development, starting from data connectivity to final report visualization. The process of connecting data from various sources helped in understanding how structured and unstructured data can be integrated into a single analytical environment. In addition, significant experience was gained in using Power Query and its M language for data transformation. This included cleaning datasets, handling missing values, transforming columns, and preparing data for analysis. These steps are critical in ensuring that the data used for visualization is accurate, consistent, and reliable. The project also involved the development of DAX (Data Analysis Expressions) measures, which played a vital role in calculating key metrics such as revenue, profit, growth percentage, and other performance indicators. Writing DAX formulas enhanced logical thinking and problem-solving skills, as it required a deep understanding of data relationships and calculation contexts.

Another major learning outcome of this project was the application of dimensional data modeling techniques, particularly the implementation of a star schema. This involved organizing data into fact and dimension tables to improve performance and maintain clarity in data relationships. The use of star schema design ensured that the data model remained scalable, efficient, and easy to understand. This practical implementation strengthened the theoretical concepts of database design and demonstrated their importance in Business Intelligence systems.

Furthermore, the project provided valuable experience in report design and visualization. Creating dashboards required careful selection of charts, layout design, color schemes, and interactive elements such as slicers and filters. This helped in developing the ability to present complex data in a simple and visually appealing manner. The importance of user-friendly design and effective data storytelling became evident during this stage.Security implementation was also an essential part of the project, where basic concepts of data access control and role-based security were explored. This ensured that sensitive information could be protected and only accessible to authorized users. In addition to technical skills, the project also contributed to the development of structured project management abilities. This included requirements gathering, planning, designing, testing, and final delivery of the solution. Each phase of the project lifecycle was executed systematically, providing hands-on experience in managing a complete BI project from start to finish.

Overall, this project has significantly enhanced both technical expertise and analytical thinking, preparing for future challenges in the field of Business Intelligence and data analytics.

5.1.4 Deviation from Expected Results

The project was largely executed as planned, with most of the defined objectives successfully achieved within the given timeline and scope. However, a few minor deviations from the initially expected outcomes were observed due to practical constraints related to time, system capabilities, and resource availability. These deviations did not significantly impact the overall functionality or effectiveness of the Healthcare Analysis Dashboard but are important to acknowledge for completeness and future improvement.

One of the primary deviations was related to mobile layout optimization, which was initially planned as part of the final deliverable. Due to limited time during the testing and refinement phase, full optimization for mobile devices was deferred to Phase 2. While the dashboard remains accessible on mobile devices, it has not been fully optimized for smaller screens, which may affect user experience for mobile-based access.

Another deviation involved the implementation of advanced AI-powered visuals, such as the Decomposition Tree. Although the feature was included in the dashboard, it was implemented in a basic configuration rather than a fully optimized and interactive version. This limitation was primarily due to time constraints and the need to prioritize core analytical features over advanced enhancements.

Additionally, the data refresh frequency was set to twice daily instead of the initially intended near real-time refresh. This adjustment was necessary due to limitations in the data source infrastructure, which does not currently support incremental refresh capabilities without a Power BI Premium license. As a result, real-time analytics could not be fully achieved within the scope of this project.

Despite these deviations, the overall system performance, analytical capabilities, and usability of the dashboard remain strong. The identified limitations provide clear direction for future enhancements, including mobile optimization, advanced AI integration, and real-time data processing. These improvements can be implemented in subsequent phases to further enhance the effectiveness and scalability of the solution..

5.2 Future Work

5.2.1 Short-Term Enhancements (Next 3 months)

- Mobile layout optimization: Design mobile-responsive report layouts for all dashboard pages to support field sales teams accessing dashboards on smartphones and tablets.
- Automated alerts and subscriptions: Configure Power BI Service data alerts and email subscriptions to proactively notify stakeholders of threshold breaches.
- Additional data sources: Integrate CRM system data to add pipeline and opportunity tracking, and website analytics data for digital channel performance monitoring.
- Paginated reports: Develop paginated report versions of key operational reports for print-quality formatted output.
- The long-term vision of this project aims to transform the Power BI dashboard into an enterprise-level Business Intelligence solution.
- The focus is on scalability, performance, and seamless integration with organizational systems.
- One major goal is to improve accessibility of dashboards across different platforms and users.

- Power BI Embedded will be implemented to integrate dashboards into internal web portals.
- This will allow users to access reports directly within their existing workflow environment.
- It will reduce dependency on switching between multiple applications for analysis.
- Another key enhancement is the adoption of Power BI Dataflows.
- Dataflows will centralize data transformation processes currently handled in Power Query.
- This will ensure consistency across multiple reports and dashboards.
- Reusability of transformation logic will reduce redundancy and improve efficiency.
- With business growth, data volume is expected to increase significantly.
- To handle this, Azure Synapse Analytics integration is proposed.
- Azure Synapse will provide a scalable and high-performance data warehousing solution.
- It will support advanced analytics and large-scale data processing.
- Real-time data processing capabilities can also be explored in the future.
- A strong BI governance framework will be established for long-term sustainability.
- This includes creating a centralized data catalog for better data management.
- Data stewardship roles will be defined to maintain data quality, security, and compliance.

5.2.2 Medium-Term Enhancements (3-12 months)

- Predictive analytics integration: Incorporate machine learning models for patient forecasting, disease trend prediction, and hospital admission rates using Python or R script visuals, or Azure Machine Learning integration. These models can help in early detection of high-risk patients, forecasting resource demand, and improving treatment planning through data-driven predictions.
- Incremental refresh: Implement incremental refresh policies to enable more frequent updates of healthcare data such as patient records, treatment logs, and hospital metrics without requiring a full dataset refresh. This will significantly improve performance

and ensure that the dashboard reflects near real-time insights while optimizing system resources.

- Natural language Q&A: Configure Power BI's Q&A natural language query feature with healthcare-specific synonyms and predefined questions. This will allow doctors and administrators to ask queries such as “number of patients with diabetes this month” or “average hospital stay by condition” in simple language, improving accessibility for non-technical users.
- Composite model: Evolve the data model to combine Import mode tables with DirectQuery connections to real-time hospital systems and databases. This hybrid approach will enable faster performance for historical data while providing live access to critical operational data such as patient admissions and emergency cases.
- Advanced visualization enhancements: Introduce more advanced and interactive visuals such as decomposition trees, key influencers, and drill-through dashboards to provide deeper insights into patient data and treatment outcomes. This will enhance analytical capabilities and support better decision-making.
- Automated alerts and notifications: Configure data-driven alerts in Power BI Service to notify healthcare administrators when critical thresholds are reached, such as sudden increases in patient admissions or abnormal test result trends. This will enable proactive response and improved operational control.
- Data quality and validation improvements: Implement automated data validation checks and monitoring mechanisms to ensure accuracy and completeness of healthcare data. This will reduce errors and improve reliability of insights generated from the dashboard.
- Integration with external healthcare systems: Enable integration with hospital management systems, laboratory systems, and other healthcare databases to ensure seamless data flow and improved data availability for analysis.
- User training and adoption programs: Conduct structured training sessions and create user guides to improve adoption among healthcare professionals. Ensuring that users understand how to interact with dashboards will maximize the value derived from the BI solution.

- Performance optimization: Continuously optimize DAX measures, data models, and visual performance to ensure fast loading times and smooth interaction, even as data volume grows.

5.2.3 Long-Term Vision (12+ months)

- The long-term vision of this Healthcare Analysis Business Intelligence project focuses on transforming the current Power BI dashboard into a scalable, enterprise-grade analytics solution capable of supporting growing healthcare data volumes, advanced analytics capabilities, and seamless integration into hospital and healthcare management systems. Over a period of 12 months and beyond, several strategic enhancements are proposed to ensure that the system remains future-ready, efficient, and aligned with evolving healthcare requirements.
- One of the key initiatives in the long-term roadmap is the implementation of Power BI Embedded. This will enable integration of dashboards directly into hospital management systems, electronic health record (EHR) platforms, and internal healthcare portals. By embedding analytics within existing workflows, doctors, administrators, and staff can access insights without switching between systems, leading to improved efficiency, faster decision-making, and higher user adoption.
- Another important enhancement involves migrating data transformation processes to Power BI Dataflows. Currently, transformations are handled within individual reports, which can lead to redundancy and inconsistencies. By centralizing these processes, the system will ensure standardized data preparation, improved data quality, and better scalability. This approach will allow multiple dashboards and reports to rely on a single, consistent source of truth, which is critical in healthcare environments where accuracy is essential.
- As healthcare data continues to grow in both volume and complexity, the integration of Azure Synapse Analytics is proposed. This cloud-based data warehousing solution will enable efficient storage and processing of large-scale patient data, medical records, and treatment histories. It will also support advanced analytics, real-time data processing, and high-performance querying, ensuring that the BI system remains responsive even under heavy data loads.

- The long-term vision also includes the incorporation of advanced analytics and artificial intelligence capabilities. Predictive analytics can be used to forecast patient admissions, identify disease trends, and optimize resource allocation. Machine learning models can assist in early disease detection, risk assessment, and personalized treatment planning. These capabilities will significantly enhance the value of the BI system by moving from descriptive analytics to predictive and prescriptive insights.
- Data security and privacy will remain a top priority in the long-term roadmap. The implementation of advanced security mechanisms such as role-based access control, data encryption, and compliance with healthcare regulations will ensure that sensitive patient information is protected. Regular audits and monitoring systems will be introduced to maintain data integrity and prevent unauthorized access.
- Another key focus area is the development of a comprehensive BI governance framework. This includes creating a centralized data catalog, defining standardized metrics, and establishing clear data ownership and stewardship roles. A formal report validation and certification process will ensure that all dashboards meet quality standards and provide accurate insights. This governance structure will enhance trust in the system and support consistent decision-making across the organization.
- Additionally, efforts will be made to improve user experience and accessibility. This includes optimizing dashboards for mobile devices, enhancing visualization techniques, and providing personalized dashboards based on user roles. Training programs and user support systems will also be implemented to ensure that healthcare professionals can effectively use the BI tools.
- Integration with real-time data sources such as IoT-enabled medical devices and hospital monitoring systems is another long-term objective. This will enable real-time tracking of patient health metrics, hospital capacity, and resource utilization, allowing for immediate response to critical situations and improving overall patient care
- Another key focus area is the development of a comprehensive BI governance framework. This includes creating a centralized data catalog, defining standardized metrics, and establishing clear data ownership and stewardship roles. A formal report validation and certification process will ensure that all dashboards meet quality standards and provide accurate insights. This governance structure will enhance trust in the system and support consistent decision-making across the organization..

- In conclusion, the long-term vision emphasizes scalability, advanced analytics, integration, security, and governance. By implementing these enhancements, the Healthcare Analysis dashboard will evolve into a robust, intelligent, and enterprise-ready analytics platform that supports strategic decision-making, improves patient outcomes, and enhances operational efficiency in the healthcare domain.

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APPENDIX I: DAX MEASURES

This appendix contains the complete DAX measure formulas implemented in the Power BI model, organized by functional category.

A.1 Base Sales Measures

Table DAX 1.1: Healthcare Measures

Measure Name	Complete DAX Formula
Total Patients	Total Patients = COUNT(Fact_PatientRecords[PatientID])
Total Treatment Cost	Total Treatment Cost = SUM(Fact_PatientRecords[TreatmentCost])
Total Admissions	Total Admissions = COUNT(Fact_PatientRecords[AdmissionID])
Average Treatment Duration	Average Treatment Duration = AVERAGE(Fact_PatientRecords[TreatmentDuration])
Patient Count	Patient Count = DISTINCTCOUNT(Fact_PatientRecords[PatientID])
Doctor Count	Doctor Count = DISTINCTCOUNT(Fact_PatientRecords[DoctorID])
Avg Treatment Cost per Patient	Avg Treatment Cost per Patient = DIVIDE([Total Treatment Cost], [Patient Count], 0)
Treatment Efficiency %	Treatment Efficiency % = DIVIDE([Total Patients], [Total Admissions], 0)
Avg Medication Usage %	Avg Medication Usage % = AVERAGE(Fact_PatientRecords[MedicationRate])
Patients per Doctor	Patients per Doctor = DIVIDE([Total Patients], [Doctor Count], 0)

A.2 Time Intelligence Measures

Table DAX 1.2: Time Measures

Measure Name	Complete DAX Formula
Patients LY	Patients LY = CALCULATE([Total Patients], SAMEPERIODLASTYEAR(Dim_Date[Date]))
Patients YoY Growth	Patients YoY Growth = DIVIDE([Total Patients] - [Patients LY], [Patients LY], BLANK())
Patients YTD	Patients YTD = TOTALYTD([Total Patients], Dim_Date[Date])
Patients PYTD	Patients PYTD = CALCULATE([Patients YTD], SAMEPERIODLASTYEAR(Dim_Date[Date]))
Patients MTD	Patients MTD = TOTALMTD([Total Patients], Dim_Date[Date])
Patients QTD	Patients QTD = TOTALQTD([Total Patients], Dim_Date[Date])
Patients Rolling 12M	Patients Rolling 12M = CALCULATE([Total Patients], DATESINPERIOD(Dim_Date[Date], LASTDATE(Dim_Date[Date]), -12, MONTH))

Measure Name	Complete DAX Formula
MoM Growth %	MoM Growth % = VAR CurrMonth = [Patients MTD] VAR PrevMonth = CALCULATE([Total Patients], PREVIOUSMONTH(Dim_Date[Date])) RETURN DIVIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())
Treatment Cost LY	Treatment Cost LY = CALCULATE([Total Treatment Cost], SAMEPERIODLASTYEAR(Dim_Date[Date]))
Treatment Cost YoY Growth	Treatment Cost YoY Growth = DIVIDE([Total Treatment Cost] - [Treatment Cost LY], [Treatment Cost LY], BLANK())

A.3 Ranking and Top N Measures

Table DAX 1.3: Ranking Measures

Measure Name	Complete DAX Formula
Disease Rank by Patients	Disease Rank by Patients = RANKX(ALL(Dim_Disease[DiseaseName]), [Total Patients], , DESC, DENSE)
Patient Rank by Visits	Patient Rank by Visits = RANKX(ALL(Dim_Patient[PatientName]), [Total Admissions], , DESC, DENSE)
Doctor Rank by Patients	Doctor Rank by Patients = RANKX(ALL(Dim_Doctor[DoctorName]), [Total Patients], , DESC, DENSE)
Hospital Rank	Hospital Rank = RANKX(ALL(Dim_Hospital[HospitalName]), [Total Patients], , DESC, DENSE)
Is Top 10 Disease	Is Top 10 Disease = IF([Disease Rank by Patients] <= 10, 1, 0)
Patient Contribution %	Patient Contribution % = DIVIDE([Total Patients], CALCULATE([Total Patients], ALL(Dim_Disease)), 0)
Cumulative Patient %	Cumulative Patient % = DIVIDE(SUMX(TOPN([Disease Rank by Patients], ALL(Dim_Disease[DiseaseName]), [Total Patients], DESC), [Total Patients]), CALCULATE([Total Patients], ALL(Dim_Disease)), 0)

A.4 Budget and Target Measures

Table DAX 1.4: Budget Measures

Measure Name	Complete DAX Formula
Total Healthcare Budget	Total Healthcare Budget = SUM(Fact_TreatmentCost[BudgetAmount])
Treatment Cost vs Budget	Treatment Cost vs Budget = [Total Treatment Cost] - [Total Healthcare Budget]
Treatment Cost vs Budget %	Treatment Cost vs Budget % = DIVIDE([Treatment Cost vs Budget], [Total Healthcare Budget], 0)
Budget Utilization %	Budget Utilization % = DIVIDE([Total Treatment Cost], [Total Healthcare Budget], 0)
Budget Traffic Light	Budget Traffic Light = IF([Budget Utilization %] >= 1, "Green", IF([Budget Utilization %] >= 0.9, "Amber", "Red"))
Remaining Budget	Remaining Budget = [Total Healthcare Budget] - [Total Treatment Cost]

APPENDIX II: POWER QUERY M CODE

This appendix documents key Power Query M language code used in the data transformation steps.

B.1 Date Dimension Generation

Table DAX 2.1: Data Measures

Step Name	M Code / Description
Source	= let StartDate = #date(2020,1,1), EndDate = #date(2024,12,31), DateList = List.Dates(StartDate, Duration.Days(EndDate - StartDate) + 1, #duration(1,0,0,0)) in DateList
ToTable	= Table.FromList(Source, Splitter.SplitByNothing(), {"Date"}, null, ExtraValues.Error)
ChangeType	= Table.TransformColumnTypes(ToTable, {{"Date", type date}})
AddDateKey	= Table.AddColumn(ChangeType, "DateKey", each Date.Year([Date]) * 10000 + Date.Month([Date]) * 100 + Date.Day([Date]), Int32.Type)
AddYear	= Table.AddColumn(AddDateKey, "Year", each Date.Year([Date]), Int32.Type)
AddQuarter	= Table.AddColumn(AddYear, "QuarterNo", each Date.QuarterOfYear([Date]), Int32.Type)
AddMonthNo	= Table.AddColumn(AddQuarter, "MonthNo", each Date.Month([Date]), Int32.Type)
AddMonthName	= Table.AddColumn(AddMonthNo, "MonthName", each Date.MonthName([Date]), type text)
AddIsWeekend	= Table.AddColumn(AddMonthName, "IsWeekend", each Date.DayOfWeek([Date], Day.Monday) >= 5, type logical)

B.2 Sales Data Transformation Steps

Table DAX 2.2: Data Transformation Steps

Step Name	Description and M Pattern
RemoveDuplicates	Table.Distinct(Source, {"PatientID", "AdmissionID"}) — removes duplicate patient admission records
FilterNulls	Table.SelectRows(RemoveDuplicates, each [PatientID] <> null and [DiseaseID] <> null) — removes null FK rows
AddDateKey	Table.AddColumn(FilterNulls, "DateKey", each Date.Year([AdmissionDate]) * 10000 + Date.Month([AdmissionDate]) * 100 + Date.Day([AdmissionDate]), Int32.Type)
AddTreatmentCost	Table.AddColumn(AddDateKey, "TreatmentCost", each [TreatmentDuration] * [CostPerDay], type number)
AddNetCost	Table.AddColumn(AddTreatmentCost, "NetCost", each [TreatmentCost] - [InsuranceCoverage], type number)

Step Name	Description and M Pattern
AddTreatmentEfficiency	Table.AddColumn(AddNetCost, "TreatmentEfficiency", each if [TreatmentCost] = 0 then 0 else ([TreatmentCost] - [NetCost]) / [TreatmentCost], type number)
RenameColumns	Table.RenameColumns(AddTreatmentEfficiency, {{"Pat_ID", "PatientID"}, {"Dis_ID", "DiseaseID"}, {"Hosp_ID", "HospitalID"}})
SelectFinalCols	Table.SelectColumns(RenameColumns, {"PatientID", "DateKey", "DiseaseID", "HospitalID", "DoctorID", "AdmissionID", "TreatmentDuration", "TreatmentCost", "MedicationCost", "InsuranceCoverage", "NetCost", "TreatmentEfficiency"})

GIT REPOSITORY LINK

The complete project has been uploaded to a public GitHub repository. The repository contains:

- Project Report (PDF)
- Power BI Dashboard file (.pbix)
- Dataset used — Superstore.csv (sourced from Kaggle)
- Dataset Link: https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
- README.md with project description and instructions

Item	Detail
GitHub Repository URL	https://github.com/SACHINYADAV1322/HEALTH-CARE-ANALYSIS
Repository Visibility	Public
Team Lead (Uploader)	Sachin Yadav(UID: 24BCS12785)
Files Included	.pbix file, project report PDF, healthcaredataset, README.md
Upload Deadline	As specified by faculty
Google Sheet Submission	Link provided by faculty — to be filled after evaluation